Thermodynamics

Lecture Notes for B.Sc. II Semester

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How to Use These Lecture Notes

These lecture notes have been prepared with the intention of helping students find the complete syllabus in one place, presented in a simple and organized manner. Each topic begins with a brief introductory discussion, followed by step-by-step derivations to build a clear conceptual understanding.

The aim of these notes is not to go into advanced or highly specialized treatments, but rather to guide students into the natural flow of physics, making the subject approachable and intuitive.

These notes are intentionally kept concise, so that they can be revised easily and used effectively during examinations as well as for building a strong foundational understanding. For students who wish to develop a deeper conceptual insight, it is strongly recommended to read these notes alongside a standard undergraduate textbook, such as Garg, Bansal, and Ghosh's *Thermal Physics*, or any other good text on thermodynamics.

If you come across any discrepancy, misprint, or unclear explanation in these notes, you are most welcome to share your feedback or queries via email at `am.phy.in@gmail.com` .

— *Amit Vishwakarma*

Contents

1 The Zeroth Law of Thermodynamics **8**
1.1 Statement . . . 8
1.2 Mathematical Formulation . . . 8
1.3 An example of Equilibrium Constraints . . . 9

2 Ideal Gas **10**
2.1 Introduction . . . 10
2.2 Ideal Gas Laws . . . 10
2.3 Ideal Gas Equation . . . 11
2.4 Work Done By an Ideal Gas . . . 12
2.4.1 Work Done by the External Agent . . . 13
2.4.2 System Perspective (Work Done by the Gas) . . . 13
2.4.3 Total Work Done . . . 13
2.5 Kinetic Theory of Ideal Gases . . . 14
2.5.1 Pressure . . . 15
2.5.2 Kinetic Energy and Temperature . . . 16
2.6 Principle of Equipartition of Energy . . . 16
2.6.1 Consistency with Kinetic Theory . . . 16
2.6.2 Analysis of Degrees of Freedom (f) . . . 16
2.7 Maxwellian Distribution of an Ideal Gas . . . 17
2.7.1 Velocity distribution . . . 19
2.7.2 Momentum . . . 21
2.7.3 Energy Distribution . . . 21
2.8 Transport phenomenon . . . 21
2.8.1 Mean free path . . . 21
2.8.2 Viscosity: The Transport of Momentum . . . 23
2.8.3 Thermal Conduction: Transport of Energy . . . 24
2.8.4 Self-Diffusion: Transport of Mass . . . 25

3 Real Gases **26**
3.1 Andrews' Experiment on CO_2 . . . 26
3.2 Van der Waals Gas Equation . . . 27
3.2.1 Correction for Finite Molecular Volume . . . 27
3.2.2 Correction for Intermolecular Attraction . . . 28
3.3 Characteristic Temperatures of a vdW gas . . . 29
3.4 Virial Equation of State . . . 32

4 First Law of Thermodynamics **33**
4.1 Point Function and Path Function . . . 33
4.2 Internal Energy: The 1st Thermodynamic Potential . . . 34
4.2.1 Joule Expansion . . . 35
4.3 The First Law of Thermodynamics . . . 35
4.3.1 Mechanical (pressure–volume) work . . . 35
4.3.2 Generalized work . . . 36
4.3.3 Chemical work and chemical potential . . . 36
4.4 Applications of the First Law of Thermodynamics . . . 37
4.4.1 Heat Capacity (C) . . . 37
4.4.2 Adiabatic Process: Derivation of $PV^\gamma = \text{constant}$ . . . 38
4.4.3 Relation Between Adiabatic and Isothermal Elasticity . . . 39

5 Second Law of Thermodynamics **41**
5.1 Heat Engine . . . 41
5.2 Refrigerator . . . 41
5.3 Second Law of Thermodynamics . . . 42
5.3.1 Violation of Clausius Statement Implies Violation of Kelvin Statement . . . 42
5.3.2 Violation of Kelvin Statement Implies Violation of Clausius Statement . . . 43
5.4 Reversible and Irreversible Processes . . . 44
5.4.1 Reversible Process . . . 44
5.4.2 Irreversible Process . . . 45
5.5 Carnot Engine . . . 45
5.5.1 Carnot's Theorem . . . 46
5.6 Temperature scale . . . 47
5.6.1 Unattainability of Absolute Zero . . . 48
5.7 Carnot Cycle . . . 48
5.8 Otto Cycle . . . 50
5.8.1 Description of Processes . . . 51
5.8.2 Efficiency of Otto Cycle . . . 52
5.9 Diesel Engine (Diesel Cycle) . . . 53
5.9.1 Thermodynamic Processes of Diesel Cycle . . . 53
5.9.2 Efficiency of Diesel Cycle . . . 55

6 Entropy **56**
6.1 Clausius Theorem . . . 56
6.2 Consequences of Clausius Theorem . . . 57
6.2.1 Reversible Cyclic Process . . . 57
6.2.2 Irreversible Cyclic Process . . . 57
6.2.3 Path Independence for Reversible Processes . . . 58
6.3 Entropy . . . 58
6.3.1 Definition of Entropy . . . 58
6.4 Entropy Change for Irreversible Processes . . . 58
6.5 Entropy Change for an Isolated System . . . 59
6.5.1 Principle of Increase of Entropy . . . 60
6.5.2 Reversible and Irreversible Processes . . . 60

6.5.3 Entropy and Equilibrium . . . 60
6.6 First Law Revisited: Entropy Representation . . . 61
6.6.1 Significance of the Entropy Formulation . . . 61

7 Thermodynamic Potentials 62
7.1 Internal Energy U . . . 62
7.1.1 Gibbs–Duhem Relation . . . 62
7.1.2 Maxwell's First Relation . . . 63
7.2 Enthalpy H . . . 64
7.2.1 Definition of Enthalpy . . . 64
7.2.2 Maxwell's Second relation . . . 64
7.3 Helmholtz Free Energy F . . . 65
7.3.1 Definition of Helmholtz Free Energy . . . 65
7.3.2 Maxwell's Third relation . . . 65
7.4 Gibbs Free Energy G . . . 66
7.4.1 Definition of Gibbs Free Energy . . . 66
7.4.2 Maxwell's Fourth relation . . . 66

8 Thermodynamic relations 68
8.1 $C_P - C_V$ Relations . . . 68
8.2 The $T\,dS$ Equations . . . 69
8.2.1 Entropy as a Function of T and V . . . 69
8.2.2 Entropy as a Function of T and P . . . 69
8.2.3 Entropy as a Function of P and V . . . 70
8.3 Clapeyron Equation . . . 70
8.3.1 Clausius–Clapeyron Equation (Liquid–Vapour Transition) . . . 72
8.4 Joule–Thomson Expansion . . . 72
8.4.1 Isoenthalpic Nature of Joule–Thomson Expansion . . . 73
8.4.2 Joule–Thomson Coefficient . . . 73
8.4.3 Ideal Gas Case . . . 74
8.4.4 Joule–Thomson Coefficient for van der Waals . . . 75
8.5 Cooling by Adiabatic Demagnetization . . . 75
8.5.1 Thermodynamics of magnetic system . . . 76
8.5.2 Magnetic Entropy Relations . . . 76
8.5.3 Expression for Cooling . . . 76

9 Third Law of Thermodynamics 78
9.1 Planck's Statement of the Third Law . . . 78
9.1.1 Mathematical Formulation . . . 78
9.1.2 Vanishing of Heat Capacity at Absolute Zero . . . 79
9.1.3 Zero Entropy of a Perfect Crystal . . . 79
9.2 Nernst Heat Theorem . . . 79
9.2.1 Unattainability of Absolute Zero . . . 80
9.3 Residual Entropy . . . 80

10 Calorimetry and Thermometry 81
10.1 Fourier's Law of Heat Conduction . . . 81
10.2 Specific Heat . . . 82

10.3 Determination of Thermal Conductivity of Good Conductors: Searle's Method . . . 82
10.3.1 Principle . . . 82
10.3.2 Theory and Derivation . . . 83
10.4 Determination of Thermal Conductivity of Bad Conductors: Lee's Disc Method . . . 83
10.5 Newton's Law of Cooling . . . 83
10.6 Thermometers . . . 84
10.7 Ohm's Law and Electrical Conduction . . . 85
10.7.1 Joule Heating and Energy Dissipation . . . 85
10.7.2 Platinum Resistance Thermometer (PRT) . . . 86
10.7.3 Thermistors . . . 86
10.8 Seebeck Effect . . . 87
10.8.1 Thermocouple emf Between Two Metals . . . 88
10.9 Peltier Effect . . . 89
10.10 Unified Theory . . . 90
10.10.1 Onsager Reciprocity Relation . . . 90

11 Black Body Radiation **92**
11.1 Basic Definitions . . . 92
11.2 Conservation of Radiant Energy . . . 93
11.3 Kirchhoff's Law of Thermal Radiation . . . 94
11.3.1 Statement of Kirchhoff's Law . . . 94
11.3.2 Implication of the Law . . . 94
11.3.3 Proof . . . 94
11.3.4 Black Body . . . 95
11.4 Wien's Displacement Law . . . 96
11.5 Stefan–Boltzmann Law . . . 96
11.6 Rayleigh–Jeans Law . . . 97
11.6.1 Derivation . . . 97
11.6.2 Ultraviolet Catastrophe . . . 98
11.7 Planck's Radiation Law . . . 99
11.7.1 Planck's Quantum Hypothesis . . . 99
11.7.2 Average Energy of an Oscillator . . . 99
11.7.3 Planck's Radiation Law . . . 100
11.7.4 Derivation of Classical Laws from Planck's Radiation Law . . . 100

A The Polytropic Process **103**

B Equation of States **104**

C Spherical Coordinate System **107**

List of Figures

1.1 paramagnetic van der waals gas 9

2.1 Cylinder-piston system: The gas is being compressed by an external force F. 12
2.2 Velocity distributions of inert gases 20

3.1 Andrew's Isotherms of CO_2 26

5.1 Heat Engine 41
5.2 Refrigerator 42
5.3 Equivalence of Anti-Clausius Engine and Anti-Kelvin Engine 42
5.4 Equivalence of Anti-Kelvin Engine and Anti-Clausius Engine 43
5.5 Carnot Engine 46
5.6 Illustrating the proof of carnot's theorem 46
5.7 Connecting Two carnot engines 47

6.1 Illustration for Clausius theorem using a Carnot engine 56

8.1 The porous experiment 73

List of Tables

2.1 Primary Transport Phenomena 23

4.1 Examples of generalized work terms 36

A.1 Polytropic process 103

Chapter 1

The Zeroth Law of Thermodynamics

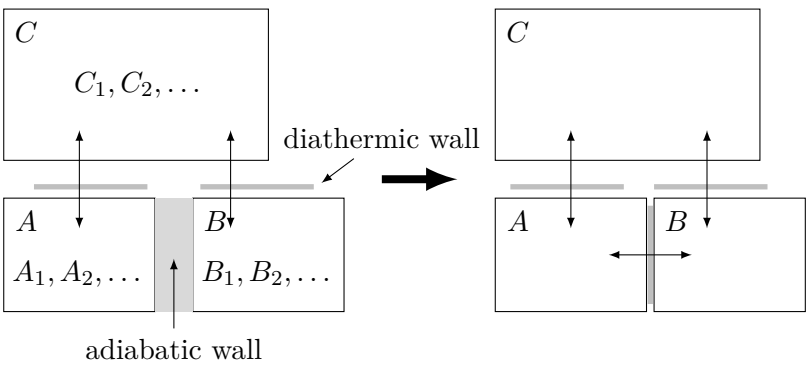
1.1 Statement

0th Law

If two systems, A and B, are separately in equilibrium with a third system, C, then they are also in equilibrium with one another.

Despite its apparent simplicity, the zeroth law has the consequence of implying the existence of an important state function, the **empirical temperature** Θ, such that systems in equilibrium are at the same temperature.

1.2 Mathematical Formulation



Let the equilibrium state of systems A, B, and C be described by the Parameter set $\{A_1, A_2, \ldots\}$, $\{B_1, B_2, \ldots\}$, and $\{C_1, C_2, \ldots\}$, respectively.

If A and C are in equilibrium implies a constraint between the coordinates of A and C. Denote this constraint by:

$$f_{AC}(A_1, A_2, \ldots; C_1, C_2, \ldots) = 0 \tag{1.1}$$

The equilibrium of B and C implies a similar constraint:

$$f_{BC}(B_1, B_2, \ldots; C_1, C_2, \ldots) = 0 \tag{1.2}$$

We can solve each of the above equations for the coordinate C_1 to yield:

$$C_1 = F_{AC}(A_1, A_2, \dots; C_2, \dots) \tag{1.3}$$
$$C_1 = F_{BC}(B_1, B_2, \dots; C_2, \dots) \tag{1.4}$$

Thus, if C is separately in equilibrium with A and B, we must have:

$$F_{AC}(A_1, A_2, \dots; C_2, \dots) = F_{BC}(B_1, B_2, \dots; C_2, \dots) \tag{1.5}$$

According to the zeroth law, there is also equilibrium between A and B, implying the constraint $f_{AB}(A_1, A_2, \dots; B_1, B_2, \dots) = 0$. This implies that it must be possible to simplify the previous equality by canceling the coordinates of C. This reduces the condition for equilibrium of A and B to:

$$\Theta_A(A_1, A_2, \dots) = \Theta_B(B_1, B_2, \dots) \tag{1.6}$$

Equilibrium is characterized by a function Θ of thermodynamic coordinates. This function specifies the *equation of state*, and *isotherms* of A are described by the condition $\Theta_A(A_1, A_2, \dots) = \Theta$.

1.3 An example of Equilibrium Constraints

We need two systems in equilibrium. For simplicity we Consider two aspects of a system (A) a paramagnetic gas of magnetization M in a magnetic field B, and (B) the same gas of volume V at pressure P.

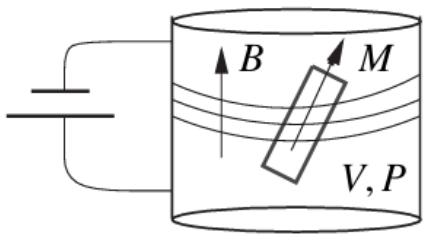


Figure 1.1: paramagnetic van der waals gas

In this system, we observe the following empirical relation similar to 1.6:

$$\frac{1}{N}\left(P + \frac{a}{V^2}\right)(V - b) \propto \frac{NB}{M} \tag{1.7}$$

this proportionality gives a constant. (Let us choose the constants that gives an easy form of the Actual Temperature)

$$\frac{1}{Nk_B}\left(P + \frac{a}{V^2}\right)(V - b) = \frac{N\mu_B^2 B}{3k_B M} = \Theta \tag{1.8}$$

These reproduce two well-known equations of state:

$$\begin{cases} (P + a/V^2)(V - b) = Nk_BT & \text{(van der Waals gas)} \\ M = (N\mu_B^2 B)/(3k_BT) & \text{(Curie paramagnet)} \end{cases} \tag{1.9}$$

where T represents the Kelvin temperature scale constructed using $\Theta = T(K)$.

Chapter 2

Ideal Gas

2.1 Introduction

An ideal gas has the following properties.

- The molecules of an ideal gas are point masses with negligible volume.
- There are no intermolecular forces between the gas molecules.
- Collisions between molecules and with the walls of the container are perfectly elastic.

2.2 Ideal Gas Laws

The behavior of an ideal gas is described by the following fundamental gas laws.

Boyle's Law

At constant temperature, the pressure of a given mass of gas is inversely proportional to its volume:

$$P \propto \frac{1}{V} \quad (T = \text{constant})$$

or

$$PV = \text{constant} \tag{2.1}$$

Charles's Law

At constant pressure, the volume of a given mass of gas is directly proportional to its absolute temperature:

$$V \propto T \quad (P = \text{constant})$$

or

$$\frac{V}{T} = \text{constant} \tag{2.2}$$

Gay–Lussac's Law

At constant volume, the pressure of a given mass of gas is directly proportional to its absolute temperature:

$$P \propto T \quad (V = \text{constant})$$

or

$$\frac{P}{T} = \text{constant} \tag{2.3}$$

Avogadro's Law

At constant temperature and pressure, the volume of a gas is directly proportional to the number of moles of the gas:

$$V \propto n \quad (T, P = \text{constant})$$

or

$$\frac{V}{n} = \text{constant} \tag{2.4}$$

This law implies that equal volumes of all gases, at the same temperature and pressure, contain the same number of molecules. As a consequence, one mole of any ideal gas occupies a volume of 22.4 L at standard temperature and pressure (STP).

2.3 Ideal Gas Equation

Above mentioned gas laws implies the following proportionality:

$$PV \propto nT \tag{2.5}$$

This gives **macroscopic equation of state** for an ideal gas is

$$\boxed{PV = nRT} \tag{2.6}$$

where R is a proportionality constant. its called the universal gas constant.

$$R = 8.314\,\mathrm{J\,mol^{-1}\,K^{-1}}$$

Fundamental Constant

The number of constituent particles in one mole of any substance is defined as **Avogadro's Number** (N_A):

$$\boxed{N_A = 6.02214076 \times 10^{23}\,\mathrm{mol^{-1}}}$$

Hence Total number of molecules of the gas of n mole is:

$$N = nN_A \tag{2.7}$$

Ideal gas equation reduces to:

$$PV = \frac{N}{N_A}RT$$

The quantity

$$\frac{R}{N_A} = k_B = 1.380649 \times 10^{-23}\,\mathrm{J\,K^{-1}} \tag{2.8}$$

is defined as the Boltzmann constant.

Thus, the ideal gas equation can be written in its **microscopic form** as

$$\boxed{PV = Nk_BT} \tag{2.9}$$

This form of the ideal gas equation relates macroscopic thermodynamic variables to microscopic molecular properties.

It leads to another form of ideal gas equation:

$$\boxed{P = \bar{n}k_BT} \tag{2.10}$$

where $\bar{n} = \frac{N}{V}$: Number density of the molecules in the system.

2.4 Work Done By an Ideal Gas

To understand the energy transfer in a thermodynamic process, we examine a gas filled inside a cylinder equipped with a movable piston.

Pressure is defined as the force acting per unit area:

$$P = \frac{F}{A} \implies F = P \cdot A \tag{2.11}$$

When an external force $\vec{F}$ is applied to the piston, causing it to move by an infinitesimal displacement $d\vec{l}$, the work done is:

$$\delta W = \vec{F} \cdot d\vec{l} \tag{2.12}$$

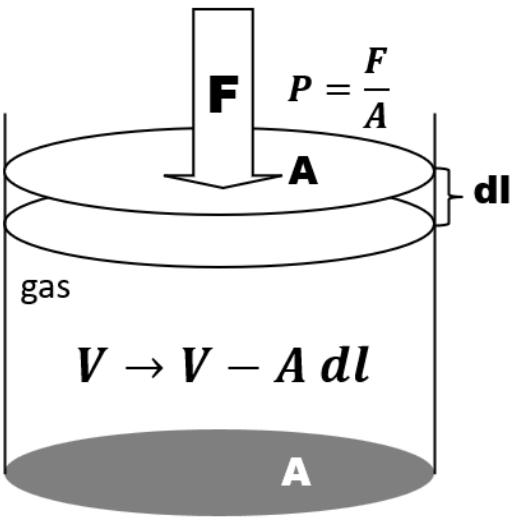


Figure 2.1: Cylinder-piston system: The gas is being compressed by an external force F.

2.4.1 Work Done by the External Agent

Substituting the force relation into the work equation, where the displacement dl occurs over the cross-sectional area A:

$$\delta W = (P \cdot A)dl \tag{2.13}$$

Since the change in volume is $dV = A \cdot dl$, the work done **by the external force on the system** is:

$$\delta W_{ext} = +P\,dV \tag{2.14}$$

2.4.2 System Perspective (Work Done by the Gas)

From the system's (the gas) point of view, if it is being compressed, the volume is decreasing. In physics, we often define work based on the system's action.

If the system is being worked upon (compressed), it is "doing" negative work because the displacement is opposite to the direction of the internal gas pressure.

Sign Convention for your Notes

- **Compression ($dV < 0$):**

  Work done *by the gas* $\delta W = PdV$ is negative ($dW < 0$).

- **Expansion ($dV > 0$):**

  Work done *by the gas* $\delta W = PdV$ is positive ($dW > 0$).

2.4.3 Total Work Done

For a finite change in volume from V_1 to V_2, the total work done by the gas is obtained by integration:

$$W = \int_{V_1}^{V_2} P\,dV \tag{2.15}$$

This expression is valid for quasi-static thermodynamic processes, where the system remains in equilibrium at every stage.

Isothermal Process

For an isothermal process ($T =$ constant) of an ideal gas ($PV = nRT$),

The work done by the gas is given by

$$W = \int_{V_1}^{V_2} \frac{nRT}{V}\,dV = nRT \ln\left(\frac{V_2}{V_1}\right) \tag{2.16}$$

Adiabatic Process

In an adiabatic process, (no heat is exchanged with the surroundings $\Delta Q = 0$). The equation of state is

$$PV^{\gamma} = \text{constant}$$

where $\gamma = \frac{C_P}{C_V}$. The work done is

$$W = \int_{V_1}^{V_2} \frac{K}{V^{\gamma}} dV = \frac{KV_1^{-\gamma}V_1 - KV_2^{-\gamma}V_2}{\gamma - 1} = \frac{P_1V_1 - P_2V_2}{\gamma - 1} \tag{2.17}$$

In this process, the work done comes at the expense of internal energy.

Isochoric Process

In an isochoric ($V =$ constant) process,

$$dV = 0$$

Hence, the work done is

$$W = \int P\,dV = 0$$

No mechanical work is done, and any heat supplied changes only the internal energy.

Isobaric Process

In an isobaric process ($P =$ constant), The work done is

$$W = \int_{V_1}^{V_2} P\,dV = P(V_2 - V_1) \tag{2.18}$$

Using the ideal gas equation ($PV = nRT$), this can also be written as

$$W = nR(T_2 - T_1) \tag{2.19}$$

Cyclic Process

In a cyclic process, the system returns to its initial state. The work done over one complete cycle is

$$W = \oint P\,dV$$

and is equal to the area enclosed by the cycle on the P–V diagram.

Free Expansion

In free expansion, the gas expands into a vacuum.

$$P_{\text{external}} = 0$$

Hence, the work done is

$$W = 0$$

No work is done despite the increase in volume.

Problem 2.1. polytropic process

calculate work done by the gas following the polytropic path: $PV^x = C$ where C is a constant. (Refer Appendix A)

2.5 Kinetic Theory of Ideal Gases

The Kinetic Theory of Gases provides a microscopic explanation for macroscopic properties like pressure and temperature. It treats a gas as a collection of a large number of molecules in constant, random motion.

Consider an ideal gas consisting of N molecules, each of mass m, enclosed in a cubical container of side L. The volume of the container is $V = L^3$.

Momentum Change per Collision

Suppose a molecule moves with velocity $\vec{v} = (v_x, v_y, v_z)$. When it collides elastically with the wall perpendicular to the x-axis, its velocity component v_x reverses to $-v_x$. The change in momentum (Δp_x) is:

$$\Delta p_x = (-mv_x) - (mv_x) = -2mv_x \tag{2.20}$$

The momentum transferred **to the wall** is $+2mv_x$.

Force exerted per collision on the wall

The time taken for the molecule to travel across the box and back to the same wall is:

$$\Delta t = \frac{2L}{v_x}$$

The rate of change of momentum (force f_x) exerted by one molecule is:

$$f_x = \frac{\Delta p}{\Delta t} = \frac{2mv_x}{2L/v_x} = \frac{mv_x^2}{L} \tag{2.21}$$

2.5.1 Pressure

The total force F_x exerted by all N molecules is:

$$F_x = \sum_{i=1}^{N} \frac{mv_{xi}^2}{L} = \frac{m}{L}\sum_{i=1}^{N} v_{xi}^2 \tag{2.22}$$

Pressure P is Force divided by Area ($A = L^2$):

$$P = \frac{F_x}{L^2} = \frac{m}{L^3}\sum v_{xi}^2 = \frac{m}{V}\sum v_{xi}^2 \tag{2.23}$$

Root Mean Square Velocity

Since the motion is random, the average velocity components are equal in all directions:

$$\langle v_x^2\rangle = \langle v_y^2\rangle = \langle v_z^2\rangle = \frac{1}{3}\langle v^2\rangle \tag{2.24}$$

Substituting $\sum v_{xi}^2 = N\langle v_x^2\rangle = \frac{1}{3}N\langle v^2\rangle$:

$$P = \frac{1}{3}\frac{Nm\langle v^2\rangle}{V} \tag{2.25}$$

$$\boxed{PV = \frac{1}{3}mN\langle v^2\rangle} \tag{2.26}$$

Defining the total mass $M = Nm$, density of the gas $\rho = M/V$ and $v_{rms} = \sqrt{\langle v^2\rangle}$:

$$\boxed{P = \frac{1}{3}\rho v_{rms}^2} \tag{2.27}$$

2.5.2 Kinetic Energy and Temperature

From the ideal gas law $PV = Nk_BT$ and 2.26:

$$\frac{1}{3}Nmv_{rms}^2 = Nk_BT \tag{2.28}$$

The average translational kinetic energy of a single molecule is:

$$\boxed{\kappa_{avg} = \langle\frac{1}{2}mv^2\rangle = \frac{1}{2}m\langle v^2\rangle = \frac{1}{2}mv_{rms}^2 = \frac{3}{2}k_BT} \tag{2.29}$$

Insight

Temperature is a direct measure of the average translational kinetic energy of the molecules.

2.6 Principle of Equipartition of Energy

Statement of the Principle

For a dynamical system in thermal equilibrium at temperature T, the total energy is distributed equally among all its active degrees of freedom. The average energy associated with each individual degree of freedom is equal to $\frac{1}{2}k_BT$.

2.6.1 Consistency with Kinetic Theory

In our previous discussion on the kinetic theory of gases, we derived the equation of state for one mole of an ideal gas as $pV = RT$. From this, we established that the average translational kinetic energy of a molecule is:

$$\bar{\kappa} = \frac{3}{2}k_BT \tag{2.30}$$

Since a monoatomic gas is considered to have only translational motion in three-dimensional space ($f = 3$), this result is perfectly consistent with the Equipartition Principle, which assigns $\frac{1}{2}k_BT$ to each of the three translational axes (x, y, z).

2.6.2 Analysis of Degrees of Freedom (f)

- **Monoatomic Gas:** As confirmed by kinetic theory, these gases possess only translational degrees of freedom ($f = 3$).

$$u_{av} = 3 \times \left(\frac{1}{2}k_BT\right) = \frac{3}{2}k_BT \tag{2.31}$$

- **Diatomic Gas:** At standard temperatures, a diatomic molecule acts as a rigid rotator. It possesses 3 translational and 2 rotational degrees of freedom (rotation about the axis joining the two atoms is neglected as the moment of inertia is negligible), giving $f = 5$.

$$u_{av} = 5 \times \left(\frac{1}{2}k_BT\right) = \frac{5}{2}k_BT \tag{2.32}$$

  unless Vibrational modes are activated.

- **Polyatomic Gas:** For more complex molecules, the situation includes:
  - **3 Translational modes.**
  - **3 Rotational modes** (or 2 if the molecule is linear).
  - **Vibrational modes:** Unlike translation or rotation, each vibrational mode contributes **two** degrees of freedom (one for kinetic energy and one for potential energy). Therefore, each active vibration contributes $1k_BT$ to the total energy.

The "Frozen" Degrees of Freedom

It is important to note that vibrational modes (and sometimes rotational ones at very low temperatures) must be "activated" by reaching a specific temperature threshold. If the thermal energy is too low, these modes remain "frozen" and do not contribute to the heat capacity of the gas.

Excercise

Given a non-linear triatomic molecule (like H_2O) where vibrations are neglected, calculate the total internal energy for 1 mole of the gas.

We will continue our discussion on various degree of freedom and associated energy with it in chapter 4

2.7 Maxwellian Distribution of an Ideal Gas

In a system of an ideal gas in thermal equilibrium at temperature T, the molecules move in random directions with varying speeds. The Maxwell-Boltzmann distribution describes the probability of a molecule having a specific velocity.

The Maxwell-Boltzmann distribution

Maxwell and Boltzmann figured out that any function of energy for an individual molecule in thermal equilibrium can be described by a specific distribution.

Discrete Form: The number of molecules n_i in a state with energy ϵ_i is proportional to the Boltzmann factor:

$$n_i \propto e^{-\frac{\epsilon_i}{k_B T}} \tag{2.33}$$

$$n_i = \frac{N}{Z} e^{-\frac{\epsilon_i}{k_B T}} \tag{2.34}$$

where $N = \sum n_i$ and $Z = \sum e^{-\epsilon_i/k_B T}$ is the partition function.
Continuous Form: For a continuous range of energies, the probability density (making sure $\int f(\epsilon) d\epsilon = 1$) is:

$$\frac{N_\epsilon d\epsilon}{N} = f(\epsilon) d\epsilon = A e^{-\frac{\epsilon}{k_B T}} d\epsilon \tag{2.35}$$

where A is a normalization constant.
$N_\epsilon d\epsilon$ is number of molecules in the range of ϵ to $\epsilon + d\epsilon$ interval.
$\frac{N_\epsilon d\epsilon}{N} = f(\epsilon) d\epsilon$ is the probability of finding that molecule (we chosen randomly) at the energy ϵ to $\epsilon + d\epsilon$ interval.

Based on the Boltzmann factor $e^{-\epsilon/k_B T}$, where $\epsilon = \frac{1}{2} m v^2$.

Out of total N molecules, Number of molecules in the velocity range from v_x to $v_x + dv_x$ will be:

$$N_{v_x} dv_x = A' \exp\left(-\frac{m v_x^2}{2 k_B T}\right) dv_x \tag{2.36}$$

Hence the probability of finding a molecule in in the velocity range from v_x to $v_x + dv_x$ will be $\frac{N_{v_x} dv_x}{N} = f(v_x) dv_x$:

$$f(v_x) dv_x = A \exp\left(-\frac{m v_x^2}{2 k_B T}\right) dv_x \tag{2.37}$$

where $A = \frac{A'}{N}$.

Normalization Constant

Since $f(v_x)$ is a probability density, it must satisfy the normalization condition

$$\int_{-\infty}^{\infty} f(v_x)\, dv_x = 1. \tag{2.38}$$

Substituting the expression for $f(v_x)$,

$$A \int_{-\infty}^{\infty} e^{-\frac{m v_x^2}{2kT}}\, dv_x = 1. \tag{2.39}$$

Using the standard Gaussian integral

$$\int_{-\infty}^{\infty} e^{-ax^2}\, dx = \sqrt{\frac{\pi}{a}}, \tag{2.40}$$

we obtain

$$\int_{-\infty}^{\infty} e^{-\frac{mv_x^2}{2kT}}\, dv_x = \sqrt{\frac{2\pi kT}{m}}. \tag{2.41}$$

which yields

$$A = \sqrt{\frac{m}{2\pi kT}}. \tag{2.42}$$

2.7.1 Velocity distribution

Substituting the value of A, the Maxwell distribution for one velocity component becomes

$$\boxed{f(v_x)dv_x = \left(\frac{m}{2\pi k_B T}\right)^{1/2} \exp\left(-\frac{mv_x^2}{2k_B T}\right) dv_x} \tag{2.43}$$

If We ask, what will be probability of randomly choosing molecule to have velocity in the (infinitesimal volume $d^3\vec{v}$) range of $\vec{v} = (v_x, v_y, v_z)$ to $\vec{v} + d^3\vec{v} = (v_x + dv_x, v_y + dv_y, v_z + dv_z)$.

Because the gas is isotropic, the motions in the x, y, and z directions are independent, meaning:

$$f(\vec{v})d^3\vec{v} = f(v_x)f(v_y)f(v_z)dv_x dv_y dv_z \tag{2.44}$$

Multiplying the three independent components together, we get the velocity vector distribution:

$$f(v_x, v_y, v_z)d^3v = \left(\frac{m}{2\pi k_B T}\right)^{3/2} \exp\left(-\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T}\right) dv_x dv_y dv_z \tag{2.45}$$

Since the energy of a molecule is purely kinetic, $\epsilon = \frac{1}{2}mv^2 = \frac{1}{2}m(v_x^2 + v_y^2 + v_z^2)$, the probability of finding a molecule with a specific velocity vector is:

$$f(\vec{v})d^3\vec{v} = \left(\frac{m}{2\pi k_B T}\right)^{3/2} \exp\left(-\frac{mv^2}{2k_B T}\right) d^3\vec{v} \tag{2.46}$$

Here volume element d^3v See Appendix

$$d^3\vec{v} = dv_x dv_y dv_z = v^2 \sin\theta d\theta d\phi \quad \text{(Spherical coordinates)} \tag{2.47}$$

Probability of finding a molecule at speed v:

$$F(v)dv = \left(\int_{\theta=0}^{\pi} \sin\theta d\theta \int_{\phi=0}^{2\pi} d\phi\right) v^2 f(v)dv \tag{2.48}$$

$$\boxed{F(v)dv = 4\pi \left(\frac{m}{2\pi k_B T}\right)^{3/2} v^2 \exp\left(-\frac{mv^2}{2k_B T}\right) dv} \tag{2.49}$$

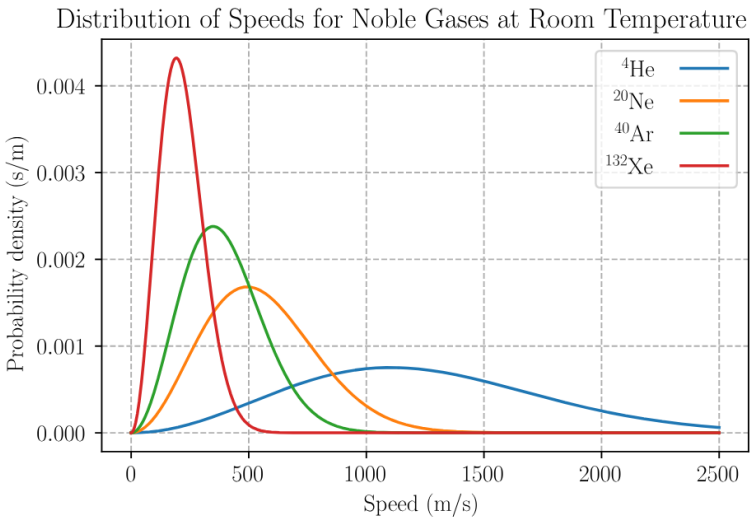


Figure 2.2: Velocity distributions of inert gases

Most Probable Speed (v_{mp})

This is found by setting $\frac{dF(v)}{dv} = 0$:

$$\frac{d}{dv}\left[v^2 \exp\left(-\frac{mv^2}{2k_BT}\right)\right] = 0 \tag{2.50}$$

yields:

$$\boxed{v_{mp} = \sqrt{\frac{2k_BT}{m}}} \tag{2.51}$$

Average Speed (v_{avg})

The average is the first moment of the distribution:

$$\langle v \rangle = \int_0^\infty vF(v)dv = \sqrt{\frac{8k_BT}{\pi m}} \tag{2.52}$$

$$\boxed{\langle v \rangle = \sqrt{\frac{8k_BT}{\pi m}}} \tag{2.53}$$

Root Mean Square Speed (v_{rms})

The mean square speed is the second moment:

$$\langle v^2 \rangle = \int_0^\infty v^2F(v)dv = \frac{3k_BT}{m} \tag{2.54}$$

$$\boxed{v_{rms} = \sqrt{\frac{3k_BT}{m}}} \tag{2.55}$$

The relationship between these speeds is:

$$\boxed{v_{mp} < v_{avg} < v_{rms}} \tag{2.56}$$

2.7.2 Momentum

Using $p = mv$, the distribution of momentum magnitude is:

$$F(p)dp = 4\pi \left(\frac{1}{2\pi m k_B T}\right)^{3/2} p^2 \exp\left(-\frac{p^2}{2mk_B T}\right) dp \tag{2.57}$$

2.7.3 Energy Distribution

The kinetic energy of a gas molecule is given by

$$\epsilon = \frac{1}{2}mv^2. \tag{2.58}$$

Using the Maxwell velocity distribution and transforming from velocity space to energy space, the distribution of translational energy is obtained as

$$F(\epsilon)\, d\epsilon = \frac{2}{\sqrt{\pi}} \frac{1}{(kT)^{3/2}} \sqrt{\epsilon}\, e^{-\epsilon/kT}\, d\epsilon. \tag{2.59}$$

Averag energy

This distribution shows that molecular energies are not equal but are spread over a range of values, with the mean energy given by

$$\langle \epsilon \rangle = \frac{3}{2}kT. \tag{2.60}$$

Using $\epsilon = \frac{1}{2}mv^2$, we transform the variable to energy:

$$F(\epsilon)d\epsilon = \frac{2\sqrt{\epsilon}}{\sqrt{\pi}(k_B T)^{3/2}} \exp\left(-\frac{\epsilon}{k_B T}\right) d\epsilon \tag{2.61}$$

2.8 Transport phenomenon

2.8.1 Mean free path

The *mean free path* (λ) is defined as the average distance traveled by a gas molecule between two successive collisions.

Collision Rate

Consider a single molecule (as hard spheres of diameter d), moving with speed v through a background of identical molecules with number density n (molecules per unit volume). A collision occurs whenever the centers of two molecules come within a distance d of each other.

During a short time interval dt, the molecule sweeps out a cylindrical volume:

$$dV = \sigma v\, dt = \pi d^2 v\, dt \tag{2.62}$$

where $\sigma = \pi d^2$ is the effective collision cross-section. The number of molecules dN within this volume (and thus the number of collisions) is:

$$dN = n(\pi d^2 v\, dt) \tag{2.63}$$

The collision frequency f is the number of collisions per unit time:

$$f = \frac{dN}{dt} = n\pi d^2 v \tag{2.64}$$

Mean Free Path

The average time between collisions, known as the *mean free time* (τ), is the reciprocal of the frequency:

$$\tau = \frac{1}{n\pi d^2 v} \tag{2.65}$$

The mean free path λ is the distance covered during this time:

$$\lambda = v\tau = \frac{1}{n\pi d^2} \tag{2.66}$$

Correction for Relative Motion (First Order Approximation)

The initial derivation assumes "target" molecules are stationary. In reality, all molecules are moving randomly. To account for this, we must use the average relative velocity v_{rel}.

For any two molecules with velocities $\vec{v}_1$ and $\vec{v}_2$, the relative velocity is $\vec{v}_{rel} = \vec{v}_1 - \vec{v}_2$. Squaring this gives:

$$v_{rel}^2 = v_1^2 + v_2^2 - 2\vec{v}_1 \cdot \vec{v}_2 \tag{2.67}$$

Taking the average over the distribution:

1. Due to random motion, the dot product term $2\vec{v}_1 \cdot \vec{v}_2$ averages to zero.
2. Since the molecules are identical, $\langle v_1^2 \rangle = \langle v_2^2 \rangle = \langle v^2 \rangle$.

Thus, $\langle v_{rel}^2 \rangle = 2\langle v^2 \rangle$, which implies $v_{rel} = \sqrt{2}v$. Substituting this into our collision frequency calculation yields the corrected mean free path:

$$\boxed{\lambda = \frac{1}{\sqrt{2}n\pi d^2}} \tag{2.68}$$

Dependence on Temperature and Pressure

Using the ideal gas law $P = nk_BT$, we substitute $n = \frac{P}{k_BT}$:

$$\lambda = \frac{k_BT}{\sqrt{2}\pi d^2 P} \tag{2.69}$$

- Typical values of the mean free path in gases at atmospheric pressure are of the order of 10^{-7} m.

Survival Equation

Number of particles surviving collisions in traveling a distance x:

$$N(x) = N(0)\exp\left(-\frac{x}{\lambda}\right) \tag{2.70}$$

Microscopic motion to Macroscopic transport

The mean free path provides the crucial link between the microscopic motion of molecules and the macroscopic transport properties of gases.

Transport Phenomena in Gases

When a gas is not in equilibrium—due to differences in temperature, velocity, or concentration—random molecular motion acts to equalize these gradients. This process is known as *transport phenomena*. Molecules "transport" physical quantities as they move through their mean free path (λ).

Table 2.1: Primary Transport Phenomena

Phenomenon	**Quantity Transported**	**Driving Gradient**	**Coefficient**
Viscosity	Momentum (mv)	Velocity (du/dz)	Viscosity (η)
Thermal Conduction	Heat Energy	Temperature (dT/dz)	Conductivity (K)
Diffusion	Mass/Molecules	Concentration (dn/dz)	Diffusion (D)

2.8.2 Viscosity: The Transport of Momentum

Viscosity is the internal friction of a gas, resulting from the transfer of momentum from high-velocity layers to low-velocity layers.

Consider a gas flowing in the x-direction with a velocity gradient du/dz along the z-axis. Consider an imaginary plane P at height z where the flow velocity is u.

- Molecules reaching P from **above** ($z + \lambda$) carry momentum: $m(u + \lambda\frac{du}{dz})$.
- Molecules reaching P from **below** ($z - \lambda$) carry momentum: $m(u - \lambda\frac{du}{dz})$.

Let n be the number density and $\bar{v}$ be the average thermal speed. Statistically, the number of molecules crossing the plane P per unit area per unit time from one side (assuming movement in six directions) is $\frac{1}{6}n\bar{v}$.

The downward momentum flux $p_\downarrow$ is:

$$p_\downarrow = \frac{1}{6}nm\bar{v}\left(u + \lambda\frac{du}{dz}\right) \tag{2.71}$$

The upward momentum flux $p_\uparrow$ is:

$$p_\uparrow = \frac{1}{6}nm\bar{v}\left(u - \lambda\frac{du}{dz}\right) \tag{2.72}$$

The net momentum transfer per unit area per unit time (shear stress or viscous force τ) is the difference:

$$\tau = p_\downarrow - p_\uparrow = \frac{1}{6}nm\bar{v}\left(2\lambda\frac{du}{dz}\right) = \frac{1}{3}nm\bar{v}\lambda\frac{du}{dz} \tag{2.73}$$

By Newton's Law of Viscosity, $\tau = \eta\frac{du}{dz}$. Comparing the expressions, we find:

$$\boxed{\eta = \frac{1}{3}nm\bar{v}\lambda} \tag{2.74}$$

Properties of Gas Viscosity

1. **Pressure Independence:** Since $\lambda = \frac{1}{\sqrt{2}\pi d^2 n}$, substituting this into η cancels n. Thus, η is independent of pressure for an ideal gas.

2. **Temperature Dependence:** Since $\bar{v} \propto \sqrt{T}$, the viscosity increases with temperature: $\eta \propto \sqrt{T}$.

2.8.3 Thermal Conduction: Transport of Energy

Thermal conduction is the process by which kinetic energy is transported from a high-temperature region to a low-temperature region.

Let ϵ be the average kinetic energy of a molecule. In a temperature gradient dT/dz, the energy of a molecule is $\epsilon = C_v T$, where c_v is the molecular heat capacity.

The energy carried by molecules crossing a plane z from the hotter side $(z + \lambda)$ and colder side $(z - \lambda)$ results in a net heat flux q:

$$q = \frac{1}{6} n\bar{v} \left(\epsilon - \lambda \frac{d\epsilon}{dz} \right) - \frac{1}{6} n\bar{v} \left(\epsilon + \lambda \frac{d\epsilon}{dz} \right) \tag{2.75}$$

$$q = -\frac{1}{3} n\bar{v}\lambda \frac{d\epsilon}{dz} \tag{2.76}$$

Since $\frac{d\epsilon}{dz} = \frac{d\epsilon}{dT}\frac{dT}{dz} = C_v \frac{dT}{dz}$:

$$q = -\left(\frac{1}{3} n\bar{v}\lambda C_v \right) \frac{dT}{dz} \tag{2.77}$$

Comparing this to Fourier's law $q = -K\frac{dT}{dz}$, we find the thermal conductivity:

$$\boxed{K = \frac{1}{3} n\bar{v}\lambda C_v} \tag{2.78}$$

properties of Thermal conductivity

- **Pressure Independence:** In both η and K, the term $n\lambda$ appears. Since $\lambda \propto 1/n$, the density cancels out, making both coefficients independent of pressure.

- **Relationship:**

$$\frac{K}{\eta} = \frac{C_v}{m} = s_v \tag{2.79}$$

$$\boxed{K = \eta m C_v} \tag{2.80}$$

It shows that, gases with high viscosity are typically also good thermal conductors.

2.8.4 Self-Diffusion: Transport of Mass

Self-diffusion refers to the transport of molecules within a gas due to a concentration gradient dn/dz.

Let n be the number density of a specific type of molecule (or labeled molecules). The number of molecules crossing a plane P per unit area per unit time is $\frac{1}{6}\bar{v}n$.

In the presence of a gradient, the net flux J is the difference between those moving up from a region of density $(n - \lambda\frac{dn}{dz})$ and those moving down from $(n + \lambda\frac{dn}{dz})$:

$$J = \frac{1}{6}\bar{v}\left(n - \lambda\frac{dn}{dz}\right) - \frac{1}{6}\bar{v}\left(n + \lambda\frac{dn}{dz}\right) \tag{2.81}$$

$$J = -\frac{1}{3}\bar{v}\lambda\frac{dn}{dz} \tag{2.82}$$

Comparing this to Fick's Law, $J = -D\frac{dn}{dz}$, we find the diffusion coefficient:

$$\boxed{D = \frac{1}{3}\bar{v}\lambda} \tag{2.83}$$

This relationship implies that as a gas becomes less dense, its molecules diffuse more rapidly.

Einstein and the Diffusion Coefficient

In 1905, **Albert Einstein** provided the first quantitative explanation of diffusion through his theory of *Brownian motion*. He showed that the diffusion coefficient is not merely a phenomenological constant but is fundamentally connected to thermal agitation and molecular mobility.

For a particle undergoing random thermal motion in a fluid, Einstein derived the relation

$$D = \mu k_{\mathrm{B}} T, \tag{2.84}$$

where μ is the mobility of the particle, k_{B} is Boltzmann's constant, and T is the absolute temperature.

This result, known as the **Einstein diffusion relation**, bridges microscopic dynamics and macroscopic transport laws. In gases, the kinetic-theory expression

$$D = \frac{1}{3}\,\bar{v}\,\lambda$$

emerges naturally as a specific realization of Einstein's general framework, with the mean free path λ and average molecular speed $\bar{v}$ encoding molecular mobility.

Einstein's work not only confirmed the atomic nature of matter but also established diffusion as a cornerstone of modern statistical mechanics.

Chapter 3

Real Gases

3.1 Andrews' Experiment on CO_2

In an attempt to liquefy CO_2 at different temperatures, Thomas Andrews conducted experiments in 1869. He applied pressure while keeping Temperature fixed, and observe how volume changes with pressure.

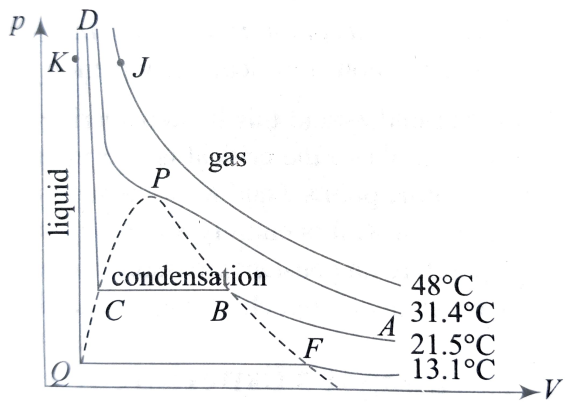


Figure 3.1: Andrew's Isotherms of CO_2

key observations:

- At high temperatures, the isotherms resemble those predicted by the ideal gas law ($PV =$ Constant, Rectangular Hyperbola).
- At lower temperatures, the isotherms exhibit a horizontal portion CB, where pressure remains constant over a range of volumes. (sudden shrink in gas volume, Condensation of gas into liquid), This horizontal region CB corresponds to the coexistence of liquid and gaseous phases.
- As the temperature increases, the horizontal portion of the isotherm shortens and finally disappears at a particular temperature (Point P) called the **critical temperature** $T_c = 31.4^0C$.

At the critical point P:

The pressure and volume at this point are known as the **critical pressure** P_c and **critical volume** V_c, respectively.

- **Critical Temperature (T_c):** The temperature above which a gas cannot be liquefied by pressure alone.
- **Critical Pressure (p_c):** The pressure required for liquefaction at T_c.
- **Critical Volume (V_c):** The volume occupied by 1 mole at p_c and T_c.

- At 13.4°C, CO_2 begins to liquefy at approximately 50 atm (Point P). During liquefaction, pressure remains constant (Line BC) until the gas is entirely liquid. The steep curve CD and QK shows that liquids are highly incompressible.
- The critical point (T_c, P_c, V_c) is a point of inflection (where curve goes concave to convex).

The critical point corresponds to a point of inflection on the critical isotherm:

$$\left(\frac{\partial P}{\partial V}\right)_T = 0, \tag{3.1}$$

$$\left(\frac{\partial^2 P}{\partial V^2}\right)_T = 0. \tag{3.2}$$

3.2 Van der Waals Gas Equation

The ideal gas equation of state assumes that gas molecules occupy negligible volume and exert no forces on one another.

In real gases, both assumptions fail. van der Waals introduced corrections to account for the finite size of molecules and intermolecular attractions.

3.2.1 Correction for Finite Molecular Volume

Real gas molecules have a finite size and therefore exclude a certain volume from being available for molecular motion. Lets calculate the volume not availble to a molecule in Total Volume V,

When two hard spherical molecules of radius r approach each other, the center of one molecule cannot come within a distance $2r$ of the center of the other. Hence, the unavailable volume around each molecule is a sphere of radius $2r$.

The non-available volume at per two molecule location is therefore

$$v'_{\text{non-available}} = \frac{4}{3}\pi(2r)^3 = 8\left(\frac{4}{3}\pi r^3\right). \tag{3.3}$$

Since this non-available volume is shared equally between two molecules, the effective non-available volume per molecule becomes

$$v_{\text{non-available}} = \frac{v'_{non-available}}{2} = 4\left(\frac{4}{3}\pi r^3\right). \tag{3.4}$$

If there are n mole (gas of N molecules, $N = nN_A$) of molecule are in total Volume V, than non-available Volume to a molecule will be

$$V_{non-available} = 4nN_A\left(\frac{4}{3}\pi r^3\right) \tag{3.5}$$

In the ideal gas scene, gas in volume V (number of moles $= n$) has a volume of V available for molecular motion. Here, the volume available for molecular motion will be reduced by $V_{non-available}$. where

$$V_{\text{available}} = V - V_{non-available}. \tag{3.6}$$

Accordingly, the volume V in the ideal gas equation must be replaced by $V - 4nN_A\left(\frac{4}{3}\pi r^3\right)$. Ideal gas equation now reduced to

$$PV_{available} = nRT \tag{3.7}$$

$$P\left[V - 4nN_A\left(\frac{4}{3}\pi r^3\right)\right] = nRT \tag{3.8}$$

or we can introduce a parameter b such that

$$P\left(V - nb\right) = nRT \tag{3.9}$$

where b is the volume not available for a molecule for motion in Volume V, that contains 1 mole of gas.

$$\boxed{b = 4N_A\left(\frac{4}{3}\pi r^3\right) = \frac{1}{n}V_{non-available}} \tag{3.10}$$

3.2.2 Correction for Intermolecular Attraction

In a real gas, attractive forces act between molecules. A molecule approaching the container wall experiences an inward pull due to neighboring molecules, resulting in a reduction of momentum transferred to the wall. Consequently, the observed pressure P is less than the pressure P_{ideal} that would be exerted in the absence of intermolecular attractions.

The magnitude of this pressure reduction depends on the number of interacting molecular pairs density. In Gas of N molecules, one molecule can make pair with $(N-1)$ other molecules. hence total of N molecules can make $N(N-1)/2 \propto N^2$ pairs.

number of interacting pairs per unit volume

$$\text{pair density} \propto \frac{N^2}{V^2} \tag{3.11}$$

Since pressure is determined by local molecular interactions near the container wall and intermolecular attraction is a pairwise effect, the reduction in pressure is proportional to the local density of interacting molecular pairs, which varies as the square of molecular density

$$\Delta P \propto \frac{N^2}{V^2} \tag{3.12}$$

$$P_{ideal} - P = \frac{an^2}{V^2} \tag{3.13}$$

where a is a constant characteristic of the strength of intermolecular attraction. P_{ideal} is given by Eq. 3.9

$$P_{ideal} = \frac{nRT}{(V - nb)} \tag{3.14}$$

By equation 3.14 and 3.14 :

$$\boxed{\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT.} \tag{3.15}$$

- The constant a measures the strength of intermolecular attraction. and Hence Larger values of a correspond to gases that are more easily liquefied.
- The constant b represents the effective molecular volume.

Johannes Diderik van der Waals

His equation of state was the first to bridge the gap between gases and liquids, establishing the principle of the "continuity of states".

Limitations of van der Waals Equation
Despite its success, the van der Waals equation has limitations:

- It fails to accurately predict behavior near the critical point.
- Quantitative agreement with experiments is limited.
- It does not account for molecular shape or quantum effects.

Nevertheless, it represents a major improvement over the ideal gas model.

3.3 Characteristic Temperatures of a vdW gas

Critical Temperature

The *critical temperature* T_c of a gas is the highest temperature above which the gas cannot be liquefied by the application of pressure alone.

For a van der Waals gas, the critical temperature is obtained from the critical point conditions

$$\left(\frac{\partial P}{\partial V}\right)_T = 0, \qquad \left(\frac{\partial^2 P}{\partial V^2}\right)_T = 0,$$

and is given by

$$T_c = \frac{8a}{27Rb}.$$

Boyle Temperature

The *Boyle temperature* T_B is the temperature at which a real gas obeys Boyle's law most closely over an extended range of pressures.

At this temperature, the second virial coefficient vanishes:

$$B(T_B) = 0.$$

For a van der Waals gas,

$$B(T) = b - \frac{a}{RT},$$

which gives

$$T_B = \frac{a}{Rb}.$$

Temperature of Inversion

The *temperature of inversion* T_i is the temperature at which the Joule–Thomson coefficient

$$\mu_{JT} = \left(\frac{\partial T}{\partial P}\right)_H$$

changes sign.

For a van der Waals gas, the Joule–Thomson coefficient is

$$\mu_{JT} = \frac{1}{C_P}\left(\frac{2a}{RT} - b\right).$$

Setting $\mu_{JT} = 0$, the (maximum) inversion temperature is

$$T_i = \frac{2a}{Rb}.$$

Relationship Between the Temperatures

Using the above expressions, we obtain

$$T_i = 2T_B, \qquad T_c = \frac{8}{27}T_B,$$

and hence

$$T_i = \frac{27}{4}T_c.$$

Important Excercise

(a) Determine the SI units for the van der Waals constants a and b based on the equation 3.15.

(b) Calculate Critical constants using Eq. 3.1 and 3.2 for Van der waals Equation of state.

Answer:

$$V_c = 3b, \tag{3.16}$$

$$P_c = \frac{a}{27b^2}, \tag{3.17}$$

$$T_c = \frac{8a}{27Rb}. \tag{3.18}$$

(c) Determine the work performed by a gas following a van der Waals equation during isothermal expansion.

(d) Using Moler Volume $V_m = \frac{V}{n}$, Find the Value of Compressibility factor $Z = \frac{PV_m}{RT}$ for the vdW Equation of state and expand in terms of Negative power of V_m.

(e) Solve $\frac{\delta Z}{\delta \rho}|_{\rho=0,T=T_B} = 0$ where $\rho = V_m^{-1}$: molar density for vdW equation of state. where T_B is known as Boyle Temperature.

answer:

$$T_B = \frac{a}{Rb} \tag{3.19}$$

(f) for dilute gas ($\rho \xrightarrow[\text{dilute gas}]{\text{high} \quad V_m} 0$), proof that vdW Equation of state reduces to:

$$PV = nRT_B \tag{3.20}$$

(g) Determine the temperature (T_i called inversion Temperature) at which the condition

$$T\left(\frac{\partial V}{\partial T}\right)_P = V$$

is satisfied.

answer:

$$T_i = \frac{2a}{Rb} \tag{3.21}$$

(h) differerence of heat capacities is given by:

$$\boxed{C_P = C_V + P\left(\frac{\partial V}{\partial T}\right)_P} \tag{3.22}$$

calculate $C_P - C_V$ for van-der waals gas.

(i) Try all the above calculations for other Equation of states given in the Appendix B

3.4 Virial Equation of State

Kamerling Onnes

The same guy famous for discovering superconductivity, Also studied the behaviour of real gases. He proposed an idea of write real gas equation as virial equation.

A more general description of real gases is provided by the virial equation of state:

$$Z = \frac{PV_m}{RT} = 1 + \frac{B(T)}{V_m} + \frac{C(T)}{V_m^2} + \cdots \tag{3.23}$$

where $B(T)$, $C(T)$, etc., are temperature-dependent virial coefficients. These coefficient represents contributions from two-body, three-body and higher-order intermolecular interactions.

since, Molar density $\rho = \frac{n}{V} = V_m^{-1}$,

Equation 3.23 takes the following form:

$$Z = \frac{P}{\rho RT} = 1 + B(T)\rho + C(T)\rho^2 + ... \tag{3.24}$$

- The second coefficient B is particularly important; it represents pairwise molecular interactions.
- The **Boyle Temperature** is defined as the temperature at which $B = 0$, where the gas obeys Boyle's law over a wide range of pressure.
- factor $Z = \frac{PV}{nRT} = \frac{PV_m}{RT} = \frac{P}{\rho RT}$ is called **compressibility factor**. For Ideal gas, Z = 1. And for real gases it deviates from 1.

Chapter 4

First Law of Thermodynamics

4.1 Point Function and Path Function

In thermodynamics, physical quantities are classified according to whether their values depend only on the state of the system or also on the manner in which the process is carried out. This leads to the concepts of *point functions* and *path functions*.

Point Function (State Function)

A **point function**, also called a *state function*, is a quantity whose value depends only on the state of the system and not on the path followed to reach that state.

If a system changes from an initial state A to a final state B, the change in a point function is uniquely determined by these two states:

$$\Delta X = X(B) - X(A), \tag{4.1}$$

and is independent of the intermediate steps of the process.

Mathematically, the differential of a point function is an *exact differential*, denoted by dX.

Examples of point functions:

Internal energy (U), Entropy (S), Enthalpy (H), Pressure (P), Volume (V), Temperature (T)

For a cyclic process,

$$\oint dX = 0, \tag{4.2}$$

which is a defining property of point functions.

Path Function

A **path function** is a quantity whose value depends on the path taken during a process and not solely on the initial and final states.

For a process from state A to state B, the value of a path function differs for different paths connecting the same states.

The differential of a path function is an *inexact differential* and is represented by symbols such as δQ or δW.

Examples of path functions:

- Heat (Q)

- Work (W)

For a cyclic process,

$$\oint \delta Q \neq 0, \qquad \oint \delta W \neq 0, \tag{4.3}$$

in general.

System possess State functions of it

Point functions represent properties stored in the system and characterize its equilibrium state. Path functions, on the other hand, represent modes of energy transfer across the system boundary during a process.
Thus, a system *possesses* internal energy or entropy, but it does not *possess* heat or work.

4.2 Internal Energy: The 1st Thermodynamic Potential

Internal energy is the total energy contained within a thermodynamic system. It is a State Function, meaning its value is determined solely by the current state of the system and not by how the system reached that state.

From a microscopic perspective, a system (such as a gas) consists of molecules in constant, random motion. The total energy of these molecules can be broken down into several components based on their Degrees of Freedom:

Translational Kinetic Energy (K_{trans}): Energy due to the linear motion of the center of mass of molecules. For the ith molecule:

$$K_{trans,i} = \frac{1}{2}mv_i^2$$

Rotational Kinetic Energy (K_{rot}): Energy associated with the rotation of molecules about their axes:

$$K_{rot,i} = \frac{1}{2}I\omega_i^2$$

Vibrational Energy (E_{vib}): Energy from the internal oscillation of atoms within a molecule (possessing both kinetic and potential components):

$$E_{vib,i} = K_{vib} + U_{vib}$$

Intermolecular Potential Energy (U_{pot}): Energy due to the forces of attraction (Van der Waals forces) or repulsion between molecules.

The **Internal Energy (U)** is the sum of all these microscopic forms of energy for all N particles in the system:

$$U = \sum_{i=1}^{N}\left(\frac{1}{2}mv_i^2 + \frac{1}{2}I\omega_i^2 + E_{vib,i}\right) + U_{potential} + ... \tag{4.4}$$

For an **Ideal Gas**, we assume that intermolecular forces are negligible ($U_{potential} \approx 0$). Thus, the internal energy of an ideal gas is purely a sum of the kinetic energies of its molecules.

4.2.1 Joule Expansion

In this experiment, a gas expands into a vacuum within a thermally insulated container. During the expansion, the Volume (V) increases and the Pressure (P) decreases. Despite the massive changes in P and V, the **Temperature (T)** of the gas remains constant (isothermal process).

Argument

Since the state changed (P and V altered) but the internal energy remained constant (as no work was done and no heat was exchanged), Joule concluded that U does not depend on volume or pressure. Instead, it is linked to the variable that did not change: Temperature.

Conclusion

For an ideal gas, the internal energy is a function of **absolute temperature only**:

$$\boxed{U = U(T)} \tag{4.5}$$

This implies that as long as the temperature of an ideal gas is constant, its internal energy remains unchanged, regardless of how the pressure or volume varies.

4.3 The First Law of Thermodynamics

The First Law of Thermodynamics is a statement of the conservation of energy.

> The heat δQ supplied to a system is used partly to increase the internal energy of the system and partly to perform work.

In finite form, the First Law is written as

$$Q = \Delta U + W \tag{4.6}$$

where W is the work done by the system.

In differential form,

$$\delta Q = dU + \delta W \tag{4.7}$$

or, equivalently, the more convenient form,

$$dU = \delta Q - \delta W \tag{4.8}$$

4.3.1 Mechanical (pressure–volume) work

If a gas exerts pressure P on its surroundings and its volume changes from V to $V + dV$, the infinitesimal work done by the system is

$$\delta W = P\,dV \tag{4.9}$$

Hence, the First Law becomes

$$\boxed{dU = \delta Q - P\,dV} \tag{4.10}$$

4.3.2 Generalized work

In a general system, work may be performed by forces other than pressure. Suppose generalized forces J_i act on the system and produce corresponding infinitesimal generalized displacements dX_i. Then the generalized work done *on* the system is

$$\delta W' = \sum_i J_i \, dX_i \tag{4.11}$$

Here, J_i are called generalized forces and X_i are the corresponding generalized displacements.

System	**Generalized Force**	**Generalized Displacement**	**Work Term**
Stretched wire	Tension F	Length dL	$F\,dL$
Liquid film	Surface tension γ	Area dA	$\gamma\,dA$
Magnetic material	Magnetic field H	Magnetization dM	$H\,dM$
Dielectric	Electric field E	Polarization dP	$E\,dP$

Table 4.1: Examples of generalized work terms

The First Law including generalized work terms may be written as

$$\delta Q + \delta W' = dU + \delta W \tag{4.12}$$

where the left-hand side represents energy supplied to the system (heat and work done on the system), while the right-hand side represents the increase in internal energy and the work done by the system.

Thus, the First Law takes the generalized form

$$dU = \delta Q - P\,dV + \sum_i J_i \, dX_i \tag{4.13}$$

We can include the term $-P\,dV$ in $J_k\,dX_k$ by identifying the pressure as a generalised force and the volume as the corresponding generalised displacement. Thus, taking

$$J_k = -P, \qquad X_k = V,$$

the work term is written as

$$\delta W = J_k\,dX_k = -P\,dV.$$

$$\boxed{dU = \delta Q + \sum_i J_i \, dX_i} \tag{4.14}$$

4.3.3 Chemical work and chemical potential

In systems where the number of particles is not conserved (such as open systems, chemical reactions, or particle exchange with a reservoir), an additional form of work appears known as *chemical work*.

If N_i denotes the number of particles of the i-th species and μ_i is the corresponding chemical potential, then the chemical work term is

$$\delta W_{\text{chem}} = -\sum_i \mu_i \, dN_i \tag{4.15}$$

The negative sign indicates that an increase in particle number increases the internal energy of the system.

Including chemical work, the most general form of the First Law becomes

$$\boxed{dU = \delta Q + \sum_i J_i \, dX_i + \sum_i \mu_i \, dN_i} \tag{4.16}$$

4.4 Applications of the First Law of Thermodynamics

4.4.1 Heat Capacity (C)

The heat capacity of a system is defined as the amount of heat required to raise the temperature of a substance containing n moles by one degree (1 K or 1°C). It is defined as

$$C = \frac{\delta Q}{dT} \qquad \Longrightarrow \qquad \delta Q = C \, dT \tag{4.17}$$

Since gases are compressible, the heat required to produce a given temperature change depends on the conditions under which heating takes place. Therefore, different heat capacities are defined depending on whether the volume or the pressure of the system is kept constant.

Heat Capacity at Constant Volume (C_V)

Consider a gas confined in a rigid container so that the volume remains constant ($dV = 0$). The work done by the system is then

$$\delta W = P \, dV = 0 \tag{4.18}$$

According to the First Law of Thermodynamics,

$$dU = \delta Q_V \tag{4.19}$$

Hence, the heat supplied at constant volume goes entirely into increasing the internal energy of the system. The heat capacity at constant volume is therefore defined by

$$\boxed{dU = C_V \, dT} \tag{4.20}$$

For an ideal gas, the internal energy U is a function of temperature only. Consequently, the relation $dU = C_V dT$ holds for an ideal gas irrespective of the nature of the process.

Heat Capacity at Constant Pressure (C_P)

If a gas is heated while maintaining constant pressure, the system is allowed to expand. In this case, the heat supplied to the system is partly used to increase the internal energy and partly to perform expansion work.

From the First Law,

$$\delta Q_P = dU + P \, dV \tag{4.21}$$

By definition,

$$\delta Q_P = C_P \, dT \tag{4.22}$$

Therefore,

$$C_P = \left(\frac{\delta Q}{\partial T}\right)_P = \left(\frac{\partial U}{\partial T}\right)_P + P\left(\frac{\partial V}{\partial T}\right)_P \tag{4.23}$$

Using the definition of C_V, this may be written as

$$\boxed{C_P = C_V + P\left(\frac{\partial V}{\partial T}\right)_P} \tag{4.24}$$

Since the system performs work during expansion at constant pressure, a greater amount of heat is required to produce the same temperature rise than in the constant volume case. Hence,

$$C_P > C_V \tag{4.25}$$

Mayer's relation

For an ideal gas obeying the equation of state $PV = nRT$,

$$\left(\frac{\partial V}{\partial T}\right)_P = \frac{nR}{P} \tag{4.26}$$

Substituting into the previous equation, we obtain Mayer's relation.

$$\boxed{C_P - C_V = nR} \tag{4.27}$$

Enthalpy: Second thermodynamic potential

we defined, Heat capacity at constant volume to be:

$$C_V = \left(\frac{\delta Q}{\delta T}\right)_V = \left(\frac{\delta U + PdV}{\delta T}\right)_V = \frac{dU}{dT} \tag{4.28}$$

For heat capacity at constant pressure:

$$C_P = \left(\frac{\delta Q}{\delta T}\right)_P = \left(\frac{dU + PdV}{\delta T}\right)_P = \frac{d(U + PV)}{dT} = \frac{dH}{dT} \tag{4.29}$$

where

$$H = U + PV \tag{4.30}$$

Known as the second thermodynamic potential, called enthalpy of the system.
For ideal gas, its only function of Temperature (similar to the Internal energy).

4.4.2 Adiabatic Process: Derivation of $PV^\gamma =$ constant

For an adiabatic process, no heat is exchanged with the surroundings:

$$\delta Q = 0 \tag{4.31}$$

The First Law of Thermodynamics,

$$dU = \delta Q - P\,dV, \tag{4.32}$$

therefore reduces to

$$dU = -P\,dV. \tag{4.33}$$

For an ideal gas, the internal energy depends only on temperature:

$$dU = C_V\,dT. \tag{4.34}$$

Hence,

$$C_V\,dT = -P\,dV. \tag{4.35}$$

The equation of state of an ideal gas is

$$PV = nRT, \tag{4.36}$$

which implies

$$P\,dV + V\,dP = nR\,dT. \tag{4.37}$$

Using the definition of heat capacities:

At constant pressure,

$$\delta Q_P = C_P\,dT = dU + P\,dV, \tag{4.38}$$

which gives

$$C_P = C_V + P\left(\frac{\partial V}{\partial T}\right)_P. \tag{4.39}$$

For an ideal gas,

$$\left(\frac{\partial V}{\partial T}\right)_P = \frac{nR}{P}, \tag{4.40}$$

and therefore Mayer's relation follows:

$$C_P - C_V = nR. \tag{4.41}$$

Combining Eqs. (4.35) and (4.37):

Eliminating dT between the two equations and using $\gamma = C_P/C_V$, we obtain

$$\frac{dP}{P} + \gamma\frac{dV}{V} = 0. \tag{4.42}$$

Integrating,

$$\boxed{PV^\gamma = \text{constant}}. \tag{4.43}$$

4.4.3 Relation Between Adiabatic and Isothermal Elasticity

For a gas, the relevant elastic constant is the *bulk modulus*, defined as the ratio of stress to strain:

$$E = \frac{\text{stress}}{\text{strain}} = -V\left(\frac{\partial P}{\partial V}\right). \tag{4.44}$$

Depending on the thermodynamic condition, two elastic moduli can be defined.

Isothermal Elasticity

For an isothermal process,

$$PV = \text{constant}. \tag{4.45}$$

Differentiating,

$$\left(\frac{\partial P}{\partial V}\right)_T = -\frac{P}{V}. \tag{4.46}$$

Hence, the isothermal elastic modulus is

$$E_T = -V\left(\frac{\partial P}{\partial V}\right)_T = P. \tag{4.47}$$

Adiabatic Elasticity

For an adiabatic process,

$$PV^\gamma = \text{constant}, \tag{4.48}$$

where $\gamma = C_P/C_V$.

Differentiating,

$$\left(\frac{\partial P}{\partial V}\right)_S = -\gamma\frac{P}{V}. \tag{4.49}$$

Thus, the adiabatic elastic modulus becomes

$$E_S = -V\left(\frac{\partial P}{\partial V}\right)_S = \gamma P. \tag{4.50}$$

Ratio of Elasticities

The ratio of adiabatic to isothermal elasticity is therefore

$$\boxed{\frac{E_S}{E_T} = \gamma = \frac{C_P}{C_V}} \tag{4.51}$$

Chapter 5

Second Law of Thermodynamics

5.1 Heat Engine

An Idealised Heat Engine works by taking in a certain amount of heat Q_H, from a heat source, converting a portion of it to work W, and dumping the remaining heat Q_c into the heat sink.

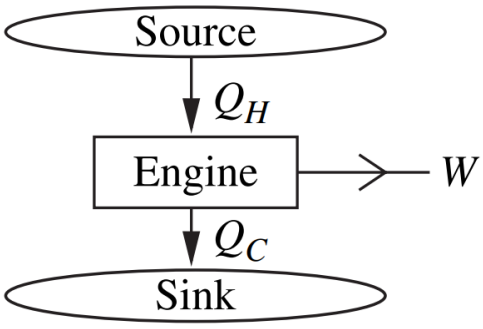


Figure 5.1: Heat Engine

Energy conservation (1st law of thermodynamics):

$$Q_H = W + Q_C \tag{5.1}$$

We Quantify the goodness of engine by how efficiently its working W by getting heat Q_H:

$$\text{Efficiency:} \quad \eta = \frac{W}{Q_H} \tag{5.2}$$

5.2 Refrigerator

An idealised refrigerator is like a heat engine running back word.

We Work W on the refrigerator, so that it can take heat Q_c from cold system, and dumping heat Q_H at a higher temperature.

Similary, from 1st law of thermodynamics:

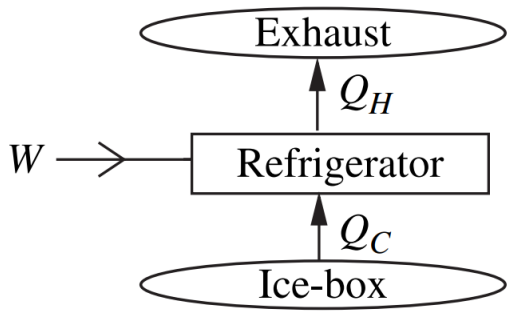


Figure 5.2: Refrigerator

$$W + Q_C = Q_H \tag{5.3}$$

We Quantify the goodness of refrigerator by its performence by taking heat from the cold system Q_C by working W on it.

$$\text{Performence:} \quad \omega = \frac{Q_C}{W} \tag{5.4}$$

5.3 Second Law of Thermodynamics

Second Law Statements

Kelvin–Planck Statement:
No process is possible whose sole result is the complete conversion of heat absorbed from a single reservoir into work.

Clausius Statement:
No process is possible whose sole result is the transfer of heat from a colder body to a hotter body without external work.

5.3.1 Violation of Clausius Statement Implies Violation of Kelvin Statement

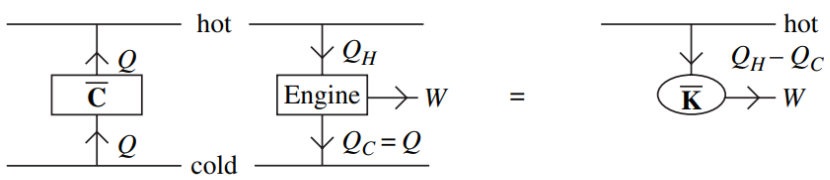


Figure 5.3: Equivalence of Anti-Clausius Engine and Anti-Kelvin Engine

Assume that the Clausius statement is violated. That is, suppose there exists an *anti-Clausius engine* which transfers an amount of heat Q from a cold reservoir to a hot reservoir without any external work.

Now consider a heat engine operating between the same two reservoirs. Let it:

- absorb heat Q_H from the hot reservoir,
- perform work W,
- reject heat $Q_C = Q$ to the cold reservoir.

For the heat engine,

$$W = Q_H - Q_C = Q_H - Q. \tag{5.5}$$

When the anti-Clausius engine and the heat engine operate together:

- The cold reservoir receives Q from the heat engine and loses Q to the anti-Clausius engine $\Rightarrow$ no net heat exchange with the cold reservoir.
- The hot reservoir supplies Q_H to the heat engine and receives Q from the anti-Clausius engine, so net heat taken from the hot reservoir is

$$Q_H - Q.$$

- The combined system produces work W.

Thus, the combined system absorbs heat $(Q_H - Q)$ from a single hot reservoir and converts it entirely into work W:

$$W = Q_H - Q. \tag{5.6}$$

This is precisely an *anti-Kelvin engine*, which violates the Kelvin–Planck statement.

Hence, violation of the Clausius statement implies violation of the Kelvin–Planck statement.

5.3.2 Violation of Kelvin Statement Implies Violation of Clausius Statement

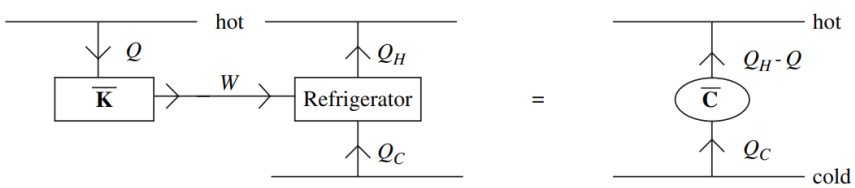


Figure 5.4: Equivalence of Anti-Kelvin Engine and Anti-Clausius Engine

Now assume that the Kelvin–Planck statement is violated. That is, suppose there exists an *anti-Kelvin engine* which absorbs heat Q from a single hot reservoir and converts it completely into work:

$$W = Q. \tag{5.7}$$

Let this work W be used to operate a refrigerator working between the same hot and cold reservoirs. Let the refrigerator:

- extract heat Q_C from the cold reservoir,
- reject heat Q_H to the hot reservoir.

For the refrigerator,

$$Q_H = Q_C + W = Q_C + Q. \tag{5.8}$$

Considering the combined system (anti-Kelvin engine + refrigerator):

- The hot reservoir loses heat Q to the anti-Kelvin engine and gains $Q_H = Q_C + Q$ from the refrigerator, so the net heat gained by the hot reservoir is Q_C.
- The cold reservoir loses heat Q_C.
- No external work is required.

Thus, the combined system transfers heat Q_C from the cold reservoir to the hot reservoir without any work input.

This constitutes an *anti-Clausius engine*, violating the Clausius statement.

Hence, violation of the Kelvin–Planck statement implies violation of the Clausius statement.

Conclusion

Since violation of either the Kelvin–Planck statement or the Clausius statement necessarily leads to violation of the other, the two statements of the second law of thermodynamics are logically equivalent.

5.4 Reversible and Irreversible Processes

The second law of thermodynamics not only restricts the direction of natural processes but also leads to an important classification of thermodynamic processes into *reversible* and *irreversible* processes.

5.4.1 Reversible Process

A thermodynamic process is said to be **reversible** if, after the process has occurred, both the system and the surroundings can be restored completely to their original states by an infinitesimal change in external conditions, leaving no net change anywhere in the universe.

In a reversible process:

- The system remains in *thermodynamic equilibrium* at every stage.
- The process proceeds infinitely slowly (quasi-static).
- There are no dissipative effects such as friction, viscosity, electrical resistance, or finite temperature differences.

A reversible process represents an *ideal limiting case* and cannot be realized exactly in practice.

Examples

- Infinitely slow isothermal expansion or compression of an ideal gas.
- Heat transfer through an infinitesimal temperature difference.
- Frictionless mechanical motion.

5.4.2 Irreversible Process

A thermodynamic process is said to be **irreversible** if the system and the surroundings cannot be restored simultaneously to their original states once the process has occurred.

In an irreversible process:

- The system passes through non-equilibrium states.
- Finite gradients (temperature, pressure, chemical potential) are present.
- Dissipative effects are unavoidable.

All natural processes occurring in finite time are irreversible.

Examples

- Free expansion of a gas into vacuum.
- Heat flow through a finite temperature difference.
- Motion with friction.
- Mixing of two gases.

Physical Origin of Irreversibility

Irreversibility arises due to:

- Friction and viscous dissipation.
- Uncontrolled heat flow.
- Finite-rate processes.
- Internal generation of entropy.

These effects convert useful work into internal energy, which cannot be fully recovered.

5.5 Carnot Engine

A carnot engine is any engine that is reversible, runs in a cycle with all of its heat exchanges taking place at a source temperature T_H and a sink at T_C.

Reversible Process

A reversible process is one that can be run backward in time by simply reversing its inputs and outputs

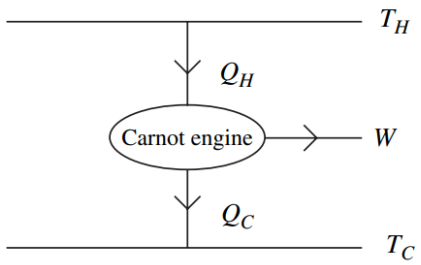


Figure 5.5: Carnot Engine

5.5.1 Carnot's Theorem

Carnot's theorem

No Engine operating between two reservoirs (at Temperature T_H and T_C) is more efficient than a Carnot engine operating between them.

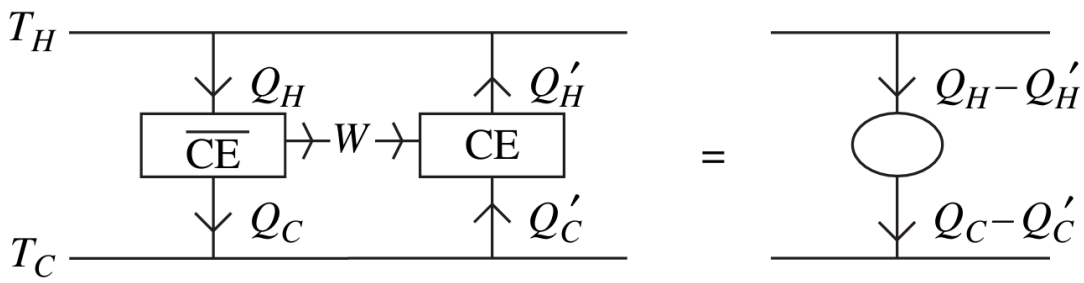


Figure 5.6: Illustrating the proof of carnot's theorem

Let say, we have a Carnot engine CE and one more engine $\bar{CE}$. Anti Carnot engine takes heat Q_H at Temperature T_H, works W and dumps Q_C at temperature T_C. (Working as a Heat Engine)

$$\eta_{\bar{CE}} = \frac{W}{Q_H} \tag{5.9}$$

Since carnot engine can run backward (reversibility), it takes Q'_C heat at temperature T_C and dump Q'_H heat at Temperature T_H. (Working as a refrigerator)

$$\eta_{CE} = \frac{W}{Q'_H} \tag{5.10}$$

This Combine system (CE + /CE) takes heat $Q_H - Q'_H$ at Temperature and dump heat $Q_C - Q'_C$ at Temperature T_C.

Energy Conservation:

$$Q_H - Q'_H = Q_C - Q'_C \tag{5.11}$$

By the Second Law of Thermodynamics:
If

$$T_H > T_C \tag{5.12}$$

Than

$$Q_H > Q'_H \quad \text{and} \quad Q_C > Q'_C \tag{5.13}$$

Which implies

$$\boxed{\eta_{CE} > \eta_{\bar{C}E}} \tag{5.14}$$

5.6 Temperature scale

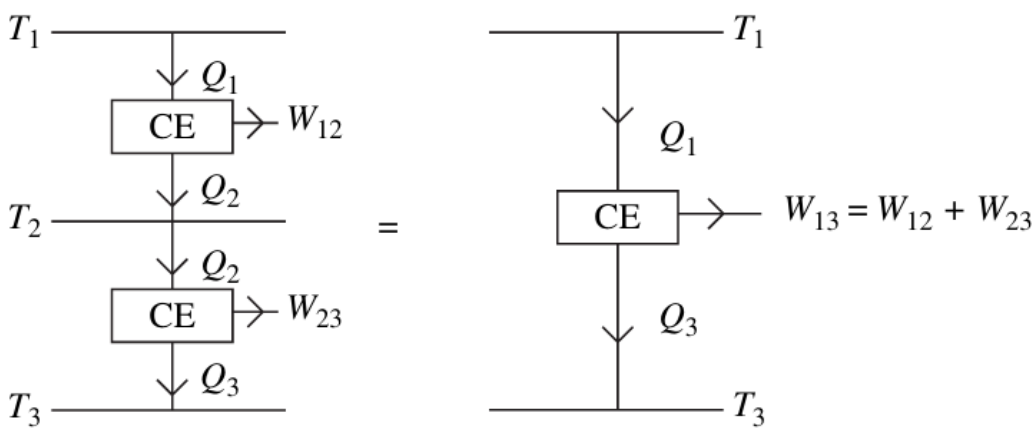


Figure 5.7: Connecting Two carnot engines

Energy conservation:

$$Q_2 = Q_1 - W_{12} = Q_1(1 - \eta(T_1, T_2)) \tag{5.15}$$

$$Q_3 = Q_2 - W_{23} = Q_2(1 - \eta(T_2, T_3)) = Q_1(1 - \eta(T_1, T_2))(1 - \eta(T_2, T_3)) \tag{5.16}$$

$$Q_3 = Q_1 - W_{13} = Q_1(1 - \eta(T_1, T_3)) \tag{5.17}$$

from 5.16 and 5.17,

$$[1 - \eta(T_1, T_2)][1 - \eta(T_2, T_3)] = [1 - \eta(T_1, T_3)] \tag{5.18}$$

To find the functional form of $\eta(T_H, T_C)$, We see from 5.18, Efficiency could be written as:

$$1 - \eta(T_1, T_2) = \frac{f(T_2)}{f(T_1)} \tag{5.19}$$

By convention, we set $\frac{f(T_2)}{f(T_1)} = \frac{T_2}{T_1}$ and hence $\eta(T_1, T_2) = 1 - \frac{T_2}{T_1}$ which implies the following:

$$\boxed{\eta(T_1, T_2) = 1 - \frac{T_2}{T_1} = 1 - \frac{Q_2}{Q_1}} \tag{5.20}$$

Or for a Carnot engine working between T_H (taking heat Q_H) and T_C (dumping heat Q_C)

$$\boxed{\eta(T_H, T_C) = 1 - \frac{T_H}{T_C} = 1 - \frac{Q_H}{Q_C}} \tag{5.21}$$

5.6.1 Unattainability of Absolute Zero

As $T_C \to 0$, $Q_C \to 0$.

Hence No heat will be exchanged with the Heat reservoir, Which implies Complete conversion of Heat Q_H into Work W. However, this contradicts the second law of thermodynamics. (Kelvin's statement)

We interpret this result to mean that Absolute zero temperature can not be achieved by any mechanical means.

In other words, the second law of thermodynamics implies unattainability of Absolute Zero Temperature.

5.7 Carnot Cycle

The Carnot cycle is an ideal reversible cycle operating between two heat reservoirs at temperatures T_H (hot reservoir) and T_C (cold reservoir), where $T_H > T_C$.

It consists of four reversible processes:

1. Isothermal Expansion at T_H
2. Adiabatic Expansion from T_H to T_C
3. Isothermal Compression at T_C
4. Adiabatic Compression from T_C to T_H

Isothermal Expansion at T_H

The gas expands isothermally from (P_1, V_1) to (P_2, V_2) at temperature T_H.

For an ideal gas:

$$P = \frac{nRT_H}{V}$$

Work done:

$$W_1 = \int_{V_1}^{V_2} P\,dV = nRT_H \int_{V_1}^{V_2} \frac{dV}{V}$$

$$W_1 = nRT_H \ln\left(\frac{V_2}{V_1}\right)$$

Heat absorbed from hot reservoir:

$$Q_1 = W_1 = nRT_H \ln\left(\frac{V_2}{V_1}\right)$$

Adiabatic Expansion

The gas expands adiabatically from (P_2, V_2, T_H) to (P_3, V_3, T_C).

For adiabatic process:

$$TV^{\gamma-1} = \text{constant}$$

No heat exchange:

$$Q_2 = 0$$

Work done:

$$W_2 = \frac{P_2V_2 - P_3V_3}{\gamma - 1}$$

Using $PV = nRT$,

$$W_2 = \frac{nR(T_H - T_C)}{\gamma - 1}$$

Isothermal Compression at T_C

The gas is compressed isothermally from (P_3, V_3) to (P_4, V_4).

Work done:

$$W_3 = \int_{V_3}^{V_4} P\,dV = nRT_C \ln\left(\frac{V_4}{V_3}\right)$$

Since $V_4 < V_3$, work is negative.

Heat rejected:

$$Q_2 = nRT_C \ln\left(\frac{V_3}{V_4}\right)$$

Adiabatic Compression

The gas is compressed adiabatically from (P_4, V_4, T_C) back to (P_1, V_1, T_H).

$$Q_4 = 0$$

Work done:

$$W_4 = \frac{nR(T_C - T_H)}{\gamma - 1}$$

Total Work Done in Carnot Cycle

Net work done:

$$W = W_1 + W_2 + W_3 + W_4$$

Since $W_2 + W_4 = 0$ (adiabatic works cancel),

$$W = W_1 + W_3$$

$$W = nRT_H \ln\left(\frac{V_2}{V_1}\right) + nRT_C \ln\left(\frac{V_4}{V_3}\right)$$

Using adiabatic relations:

$$\frac{V_2}{V_1} = \frac{V_3}{V_4}$$

Thus,

$$W = nR(T_H - T_C) \ln\left(\frac{V_2}{V_1}\right)$$

Efficiency of Carnot Engine

Efficiency is defined as:

$$\eta = \frac{\text{Net Work}}{\text{Heat Absorbed}} = \frac{W}{Q_1}$$

Substituting:

$$\eta = \frac{nR(T_H - T_C) \ln\left(\frac{V_2}{V_1}\right)}{nRT_H \ln\left(\frac{V_2}{V_1}\right)}$$

$$\boxed{\eta = 1 - \frac{T_H}{T_C}}$$

Efficiency of Carnot Engine

The efficiency of a Carnot engine depends only on the reservoir temperatures and is independent of the working substance.

$$\boxed{\eta_{\text{Carnot}} = 1 - \frac{T_H}{T_C}}$$

5.8 Otto Cycle

The Otto cycle is the ideal thermodynamic cycle for a spark–ignition (petrol) engine. It consists of four internally reversible processes:

1. Adiabatic compression

2. Constant-volume heat addition
3. Adiabatic expansion
4. Constant-volume heat rejection

The working substance is assumed to be an ideal gas. Let the compression ratio be

$$r = \frac{V_1}{V_2}$$

where V_1 is the maximum volume and V_2 is the minimum volume. Let $\gamma = \frac{C_P}{C_V}$.

Working Principle

In an Otto engine, the air–fuel mixture is first compressed adiabatically, which raises its temperature and pressure. At minimum volume, combustion occurs rapidly and heat is added at constant volume. This increases the pressure sharply. The high-pressure gas then expands adiabatically, producing useful mechanical work. Finally, heat is rejected at constant volume, returning the system to its initial state.

5.8.1 Description of Processes

Process 1 → 2 : Adiabatic Compression

$$P_1 V_1^{\gamma} = P_2 V_2^{\gamma}$$

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

Since $r = \frac{V_1}{V_2}$,

$$T_2 = T_1 r^{\gamma-1}$$

Work done during compression:

$$W_{12} = \frac{P_2 V_2 - P_1 V_1}{1-\gamma} = \frac{nR(T_2 - T_1)}{1-\gamma}$$

Process 2 → 3 : Constant Volume Heat Addition

$$V_2 = V_3$$

Work done:

$$W_{23} = 0$$

Heat supplied:

$$Q_{in} = nC_V(T_3 - T_2)$$

Pressure increases according to

$$\frac{P_3}{P_2} = \frac{T_3}{T_2}$$

Process 3 → 4 : Adiabatic Expansion

$$P_3V_3^\gamma = P_4V_4^\gamma$$

$$T_3V_3^{\gamma-1} = T_4V_4^{\gamma-1}$$

Since $V_4 = V_1$ and $V_3 = V_2$,

$$T_4 = T_3r^{1-\gamma}$$

Work done by gas:

$$W_{34} = \frac{P_4V_4 - P_3V_3}{1-\gamma} = \frac{nR(T_4 - T_3)}{1-\gamma}$$

Process 4 → 1 : Constant Volume Heat Rejection

$$V_4 = V_1$$

Work done:

$$W_{41} = 0$$

Heat rejected:

$$Q_{out} = nC_V(T_4 - T_1)$$

Net Work Done

Net work over the complete cycle:

$$W = Q_{in} - Q_{out}$$

$$W = nC_V[(T_3 - T_2) - (T_4 - T_1)]$$

5.8.2 Efficiency of Otto Cycle

Efficiency is defined as

$$\eta = \frac{W}{Q_{in}} = 1 - \frac{Q_{out}}{Q_{in}}$$

$$\eta = 1 - \frac{nC_V(T_4 - T_1)}{nC_V(T_3 - T_2)}$$

$$\eta = 1 - \frac{T_4 - T_1}{T_3 - T_2}$$

Using the adiabatic relations

$$T_2 = T_1r^{\gamma-1}$$

$$T_4 = T_3r^{1-\gamma}$$

After simplification,

$$\boxed{\eta_{\text{Otto}} = 1 - \frac{1}{r^{\gamma-1}}}$$

Efficiency of Otto Engine

The efficiency of the Otto cycle depends only on the compression ratio r and the heat capacity ratio γ, and is independent of the amount of heat supplied.

5.9 Diesel Engine (Diesel Cycle)

A diesel engine is an internal combustion engine in which ignition of fuel occurs due to the high temperature produced by compressing air inside the cylinder. Unlike petrol engines, no spark plug is required. The ideal thermodynamic cycle representing the working of a diesel engine is called the **Diesel cycle**.

The cycle consists of four processes:

1. Adiabatic compression
2. Constant pressure heat addition
3. Adiabatic expansion
4. Constant volume heat rejection

Let

$$r = \frac{V_1}{V_2} \quad \text{(compression ratio)}, \qquad \rho = \frac{V_3}{V_2} \quad \text{(cut-off ratio)}$$

and $\gamma = \dfrac{C_p}{C_v}$.

5.9.1 Thermodynamic Processes of Diesel Cycle

Process 1 → 2 : Adiabatic Compression

Air is compressed adiabatically from volume V_1 to V_2.

$$T_2 = T_1 r^{\gamma-1}, \qquad P_2 = P_1 r^{\gamma}$$

Work done during adiabatic process:

$$W_{12} = \frac{P_2V_2 - P_1V_1}{1-\gamma} = \frac{nR(T_1 - T_2)}{\gamma - 1}$$

$$W_{12} = \frac{nRT_1}{\gamma - 1}\left(1 - r^{\gamma-1}\right)$$

This work is negative (work done on the gas).

Process 2 → 3 : Constant Pressure Heat Addition

Fuel is injected and combustion occurs at constant pressure.

$$\frac{T_3}{T_2} = \frac{V_3}{V_2} = \rho \quad \Rightarrow \quad T_3 = T_2\rho$$

Work done at constant pressure:

$$W_{23} = P(V_3 - V_2) = nR(T_3 - T_2)$$

$$W_{23} = nRT_2(\rho - 1)$$

Using $T_2 = T_1 r^{\gamma-1}$:

$$W_{23} = nRT_1 r^{\gamma-1}(\rho - 1)$$

This work is positive.
Heat supplied:

$$Q_{in} = nC_p(T_3 - T_2)$$

Process 3 → 4 : Adiabatic Expansion

Gas expands adiabatically from V_3 to $V_4 = V_1$.

$$T_3 = T_1 r^{\gamma-1}\rho$$

$$T_4 = T_3\left(\frac{V_3}{V_4}\right)^{\gamma-1} = T_3\left(\frac{\rho}{r}\right)^{\gamma-1} = T_1\rho^{\gamma}$$

Work done:

$$W_{34} = \frac{nR(T_3 - T_4)}{\gamma - 1}$$

$$W_{34} = \frac{nRT_1}{\gamma - 1}\left(r^{\gamma-1}\rho - \rho^{\gamma}\right)$$

This is positive work (power stroke).

Process 4 → 1 : Constant Volume Heat Rejection

Heat is rejected at constant volume.

$$W_{41} = 0$$

Heat rejected:

$$Q_{out} = nC_v(T_4 - T_1)$$

Net Work Done in Diesel Cycle

$$W_{net} = W_{12} + W_{23} + W_{34}$$

$$W_{net} = \frac{nRT_1}{\gamma - 1}(1 - r^{\gamma-1}) + nRT_1 r^{\gamma-1}(\rho - 1) + \frac{nRT_1}{\gamma - 1}(r^{\gamma-1}\rho - \rho^{\gamma})$$

The net work done is equal to the area enclosed by the PV diagram.

5.9.2 Efficiency of Diesel Cycle

Thermal efficiency is

$$\eta = 1 - \frac{Q_{out}}{Q_{in}}$$

$$Q_{in} = nC_p(T_3 - T_2), \qquad Q_{out} = nC_v(T_4 - T_1)$$

After simplification,

$$\boxed{\eta = 1 - \frac{1}{r^{\gamma-1}}\left(\frac{\rho^{\gamma} - 1}{\gamma(\rho - 1)}\right)}$$

Chapter 6

Entropy

6.1 Clausius Theorem

Clausius Theorem

For any cyclic transformation (reversible or irreversible),

$$\oint \frac{\delta Q}{T} \leq 0 \tag{6.1}$$

where δQ is the infinitesimal heat supplied to the system at absolute temperature T.

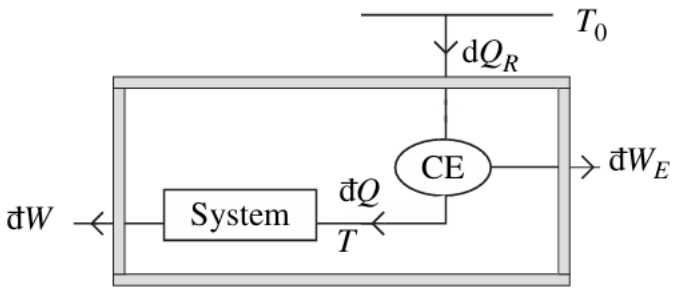


Figure 6.1: Illustration for Clausius theorem using a Carnot engine

Let us consider a Carnot engine working between two reservoirs at temperatures T_0 and T. The Carnot engine absorbs heat δQ_R from the reservoir at temperature T_0 and rejects heat δQ to a system at temperature T. Since the Carnot engine is reversible, it performs work δW_E.

For a reversible Carnot engine operating between temperatures T_0 and T,

$$\frac{\delta Q_R}{\delta Q} = \frac{T_0}{T}. \tag{6.2}$$

Hence,

$$\delta Q_R = T_0 \frac{\delta Q}{T}. \tag{6.3}$$

The system at temperature T absorbs heat δQ from the Carnot engine and performs work δW.

Over a complete cycle, the total heat absorbed by the Carnot engine is

$$Q_R = T_0 \oint \frac{\delta Q}{T}. \tag{6.4}$$

Now consider the *combined system*, consisting of the original system and the Carnot engine together. This combined system absorbs heat Q_R from the single reservoir at temperature T_0 and converts it entirely into work W', where

$$Q_R = W'. \tag{6.5}$$

This appears to violate the second law of thermodynamics (Kelvin–Planck statement), which forbids the complete conversion of heat from a single reservoir into work.

To ensure consistency with the second law, we must have

$$Q_R \leq 0. \tag{6.6}$$

Using Eqs. (6.4) and (6.6), and noting that $T_0 > 0$, we obtain

$$\boxed{\oint \frac{\delta Q}{T} \leq 0} \tag{6.7}$$

This proves Clausius theorem.

6.2 Consequences of Clausius Theorem

Clausius theorem states that for any cyclic process,

$$\oint \frac{\delta Q}{T} \leq 0. \tag{6.8}$$

This result has several important consequences.

6.2.1 Reversible Cyclic Process

For a *reversible* cyclic process, there is no dissipation and no entropy production. Hence the inequality becomes an equality:

$$\oint \frac{\delta Q_{\text{rev}}}{T} = 0. \tag{6.9}$$

This result is crucial, because it suggests that for reversible processes the quantity $\delta Q_{\text{rev}}/T$ behaves in a special way.

6.2.2 Irreversible Cyclic Process

For an *irreversible* cyclic process, dissipation is always present (for example due to friction, finite temperature gradients, etc.). In this case,

$$\oint \frac{\delta Q}{T} < 0. \tag{6.10}$$

Thus, the inequality in Clausius theorem distinguishes between reversible and irreversible processes.

6.2.3 Path Independence for Reversible Processes

Consider two equilibrium states A and B, connected by two different reversible paths R_1 and R_2. Form a closed cycle by going from A to B along R_1 and returning from B to A along R_2 (in reverse direction).

Applying Eq. (6.9) to this reversible cycle,

$$\oint \frac{\delta Q_{\text{rev}}}{T} = \int_A^B \frac{\delta Q_{\text{rev}}}{T}\bigg|_{R_1} + \int_B^A \frac{\delta Q_{\text{rev}}}{T}\bigg|_{R_2} = 0. \tag{6.11}$$

Therefore,

$$\int_A^B \frac{\delta Q_{\text{rev}}}{T}\bigg|_{R_1} = \int_A^B \frac{\delta Q_{\text{rev}}}{T}\bigg|_{R_2}. \tag{6.12}$$

This shows that for reversible processes, the integral $\int \delta Q_{\text{rev}}/T$ depends only on the initial and final states, and not on the path.

6.3 Entropy

The result obtained above suggests the existence of a new thermodynamic state function.

Since the integral

$$\int_A^B \frac{\delta Q_{\text{rev}}}{T} \tag{6.13}$$

has the same value for all reversible paths connecting two equilibrium states A and B, it must correspond to the change of some state function.

This state function is called *entropy*.

6.3.1 Definition of Entropy

Entropy, denoted by S, is defined such that its infinitesimal change is

$$dS \equiv \frac{\delta Q_{\text{rev}}}{T}. \tag{6.14}$$

For a finite reversible process between states A and B,

$$S(B) - S(A) = \int_A^B \frac{\delta Q_{\text{rev}}}{T}. \tag{6.15}$$

Thus, entropy is a *state function*, depending only on the equilibrium state of the system.

6.4 Entropy Change for Irreversible Processes

Consider an irreversible process that takes a system from an equilibrium state A to another equilibrium state B. Let the system return from B to A along a reversible path.

Together, these two processes form a closed thermodynamic cycle. According to Clausius theorem, for any cyclic process,

$$\oint \frac{\delta Q}{T} \leq 0. \tag{6.16}$$

The cyclic integral may be expressed as the sum of the integrals over the two paths:

$$\oint \frac{\delta Q}{T} = \int_A^B \frac{\delta Q}{T} + \int_B^A \frac{\delta Q_{\text{rev}}}{T}. \tag{6.17}$$

Since reversing the limits of integration changes the sign,

$$\int_B^A \frac{\delta Q_{\text{rev}}}{T} = -\int_A^B \frac{\delta Q_{\text{rev}}}{T}, \tag{6.18}$$

we obtain

$$\int_A^B \frac{\delta Q}{T} \leq \int_A^B \frac{\delta Q_{\text{rev}}}{T}. \tag{6.19}$$

The equality holds only if the entire cycle is reversible. Since the path from A to B is irreversible, the inequality is strict.

Using the definition of entropy,

$$S(B) - S(A) = \int_A^B \frac{\delta Q_{\text{rev}}}{T}, \tag{6.20}$$

we finally obtain

$$\int_A^B \frac{\delta Q}{T} < S(B) - S(A). \tag{6.21}$$

In differential form, this may be written as

$$\boxed{dS > \frac{\delta Q}{T}} \tag{6.22}$$

This inequality provides a general mathematical statement of the second law of thermodynamics for irreversible processes.

6.5 Entropy Change for an Isolated System

An isolated system does not exchange heat with its surroundings. Hence,

$$\delta Q = 0. \tag{6.23}$$

From the entropy inequality,

$$dS \geq \frac{\delta Q}{T}, \tag{6.24}$$

it follows immediately that

$$dS \geq 0. \tag{6.25}$$

Thus, the entropy of an isolated system can never decrease. It remains constant for reversible processes and increases for irreversible processes.

This result introduces a natural direction (or *arrow*) to thermodynamic processes.

6.5.1 Principle of Increase of Entropy

The behavior of entropy in isolated systems leads to a fundamental statement of the second law of thermodynamics.

Principle of Increase of Entropy

The entropy of an isolated system can never decrease:

$$\Delta S \geq 0. \tag{6.26}$$

The equality holds for reversible processes, while the inequality holds for irreversible processes.

6.5.2 Reversible and Irreversible Processes

- **Reversible process:** For a reversible process occurring in an isolated system,

$$\Delta S = 0. \tag{6.27}$$

  The entropy of the system remains constant.

- **Irreversible process:** For an irreversible process occurring in an isolated system,

$$\Delta S > 0. \tag{6.28}$$

  The entropy of the system increases.

Hence, entropy production provides a quantitative measure of irreversibility in a thermodynamic process.

Arrow of Time

The principle of increase of entropy introduces a preferred direction to physical processes. While the laws of mechanics are time-reversible, thermodynamic processes are not.
Spontaneous processes in an isolated system always proceed in the direction of increasing entropy.

6.5.3 Entropy and Equilibrium

For an isolated system in thermodynamic equilibrium,

$$dS = 0. \tag{6.29}$$

Any small spontaneous deviation from equilibrium leads to a decrease of entropy,

$$d^2S < 0. \tag{6.30}$$

Therefore, the equilibrium state of an isolated system corresponds to a state of *maximum entropy*.

Maximum Entropy Principle

Thermodynamic equilibrium in an isolated system is characterized by maximum entropy. This principle provides the final criterion for equilibrium and stability.

6.6 First Law Revisited: Entropy Representation

We now return to the first law of thermodynamics and reformulate it using the concept of entropy.

The first law for an infinitesimal process is

$$\delta Q = dU + \delta W. \tag{6.31}$$

For a reversible process, the heat absorbed by the system can be written in terms of entropy as

$$\delta Q_{\text{rev}} = T\,dS. \tag{6.32}$$

Substituting this into the first law, we obtain

$$T\,dS = dU + \delta W_{\text{rev}}. \tag{6.33}$$

For a simple compressible system performing only $P\,dV$ work,

$$\delta W_{\text{rev}} = P\,dV. \tag{6.34}$$

Hence, the first law becomes

$$\boxed{dU = T\,dS - P\,dV} \tag{6.35}$$

This equation is the *fundamental thermodynamic relation* for a simple system.

6.6.1 Significance of the Entropy Formulation

The introduction of entropy allows heat, which is a path-dependent quantity, to be expressed in terms of exact differentials.

While δQ is not a state function, the quantity $\delta Q_{\text{rev}}/T$ is an exact differential:

$$dS = \frac{\delta Q_{\text{rev}}}{T}. \tag{6.36}$$

Thus, entropy provides a bridge between energy conservation (first law) and irreversibility (second law).

Unified View

The equation

$$dU = T\,dS - P\,dV$$

combines the first and second laws into a single mathematical statement.
Energy conservation determines *how much* energy changes, while entropy determines *how* the process can occur.

- The term $T\,dS$ represents the reversible heat exchange.
- The term $P\,dV$ represents mechanical work.

Chapter 7

Thermodynamic Potentials

7.1 Internal Energy U

As introduced in Chapter 4, the internal energy U represents the total energy of a thermodynamic system.

In the most general form, the first law of thermodynamics may be written as

$$dU = T\,dS + J \cdot dx + \mu\,dN, \tag{7.1}$$

where S is the entropy, x is a generalized displacement with conjugate generalized force J, and N is the number of particles with chemical potential μ.

Equation (7.1) implies that the internal energy is a function of the extensive variables S, x, and N:

$$U = U(S, x, N). \tag{7.2}$$

From Eq. (7.1), the corresponding thermodynamic derivatives are

$$\left(\frac{\partial U}{\partial S}\right)_{x,N} = T, \qquad \left(\frac{\partial U}{\partial x}\right)_{S,N} = J, \qquad \left(\frac{\partial U}{\partial N}\right)_{S,x} = \mu. \tag{7.3}$$

For fixed intensive variables (T, J, μ), all extensive quantities are proportional to the size of the system.

7.1.1 Gibbs–Duhem Relation

Internal energy is an extensive quantity. Therefore, if the size of the system is increased by a factor λ, all extensive variables scale by the same factor:

$$U(\lambda S, \lambda x, \lambda N) = \lambda U(S, x, N). \tag{7.4}$$

Differentiating both sides with respect to λ, we obtain

$$\frac{\partial U}{\partial(\lambda S)}\frac{\partial(\lambda S)}{\partial \lambda} + \frac{\partial U}{\partial(\lambda x)}\frac{\partial(\lambda x)}{\partial \lambda} + \frac{\partial U}{\partial(\lambda N)}\frac{\partial(\lambda N)}{\partial \lambda} = U(S, x, N). \tag{7.5}$$

Using

$$\frac{\partial U}{\partial S} = T, \qquad \frac{\partial U}{\partial x} = J, \qquad \frac{\partial U}{\partial N} = \mu,$$

and setting $\lambda = 1$, we obtain

$$\boxed{U = TS + J \cdot x + \mu \cdot N} \tag{7.6}$$

Equation (7.6) shows that

Internal Energy

the internal energy consists of three contributions: thermal energy TS, mechanical work energy $J \cdot x$, and chemical potential energy $\mu \cdot N$.

Taking the total differential of Eq. (7.6), we obtain

$$dU = T\,dS + S\,dT + J \cdot dx + x \cdot dJ + \mu\,dN + N \cdot d\mu. \tag{7.7}$$

Comparing Eq. (7.7) with the first law in Eq. (7.1), we obtain

$$\boxed{S\,dT + x \cdot dJ + N \cdot d\mu = 0} \tag{7.8}$$

This relation is known as the *Gibbs–Duhem relation*. It expresses the dependence between the intensive variables T, J, and μ, and is a direct consequence of the extensivity of internal energy.

7.1.2 Maxwell's First Relation

For simplicity, we now consider a simple compressible system:

$$\boxed{dU = T\,dS - P\,dV} \tag{7.9}$$

Hence, internal energy is a function of entropy and volume:

$$U = U(S, V). \tag{7.10}$$

From this relation,

$$T = \left(\frac{\partial U}{\partial S}\right)_V, \qquad -P = \left(\frac{\partial U}{\partial V}\right)_S. \tag{7.11}$$

Since $U(S, V)$ is a well-defined state function, the mixed second derivatives must be equal:

$$\left(\frac{\partial^2 U}{\partial V \partial S}\right)_{S,V} = \left[\frac{\partial}{\partial V}\left(\frac{\partial U}{\partial S}\right)_V\right]_S = \left[\frac{\partial}{\partial S}\left(\frac{\partial U}{\partial V}\right)_S\right]_V \tag{7.12}$$

Substituting the thermodynamic derivatives, we obtain

$$\boxed{\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V} \tag{7.13}$$

This equation is known as the *first Maxwell relation*. It follows directly from the exactness of the differential dU and is independent of the microscopic nature of the system.

7.2 Enthalpy H

For an isolated system, the entropy must increase during any spontaneous process and reaches a maximum in equilibrium. This provides the equilibrium criterion when no exchange with the surroundings is allowed.

However, many physical systems are *not* isolated. They may exchange mechanical work with their surroundings, such as pressure–volume work. In such situations, entropy alone is no longer the most convenient quantity for describing equilibrium.

Consider a system that can perform mechanical work of the form $J \cdot dx$ (for example, pressure–volume work $P\,dV$) while remaining thermally insulated ($\delta Q = 0$).

For such systems, it is useful to introduce a new thermodynamic potential that naturally incorporates mechanical work. This potential is called *enthalpy*.

7.2.1 Definition of Enthalpy

Enthalpy H is defined as

$$H = U - J \cdot x. \tag{7.14}$$

For the simple compressible systems of interest, where

$$J = -P, \qquad x = V,$$

the enthalpy becomes

$$\boxed{H = U + PV} \tag{7.15}$$

Enthalpy

Thus, enthalpy represents the total energy required to create the system and make room for it against an external pressure.

7.2.2 Maxwell's Second relation

Taking the total differential of $H = U + PV$, we obtain

$$dH = dU + P\,dV + V\,dP. \tag{7.16}$$

Using the fundamental thermodynamic relation

$$dU = T\,dS - P\,dV, \tag{7.17}$$

we substitute into the above equation to obtain

$$\boxed{dH = T\,dS + V\,dP} \tag{7.18}$$

This expression shows that enthalpy is naturally a function of entropy and pressure:

$$H = H(S, P). \tag{7.19}$$

From the differential form of enthalpy,

$$dH = T\,dS + V\,dP, \tag{7.20}$$

we identify

$$T = \left(\frac{\partial H}{\partial S}\right)_P, \qquad V = \left(\frac{\partial H}{\partial P}\right)_S. \tag{7.21}$$

Since $H(S, P)$ is a state function, its mixed second derivatives must be equal:

$$\frac{\partial}{\partial P}\left(\frac{\partial H}{\partial S}\right)_P = \frac{\partial}{\partial S}\left(\frac{\partial H}{\partial P}\right)_S. \tag{7.22}$$

Substituting the thermodynamic derivatives, we obtain

$$\boxed{\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P} \tag{7.23}$$

This equation is known as the *second Maxwell relation.*

7.3 Helmholtz Free Energy F

Helmholtz free energy is the thermodynamic potential that is most useful for systems in contact with a heat reservoir at constant temperature T and fixed volume V.

In such cases, the natural control variables are T and V, and the equilibrium state corresponds to minimum Helmholtz free energy.

Helmholtz free Energy

Physically, F represents the amount of energy available to do useful work at constant temperature.

7.3.1 Definition of Helmholtz Free Energy

Helmholtz free energy is defined as

$$\boxed{F = U - TS} \tag{7.24}$$

This definition subtracts the energy "unavailable for work" due to entropy.

7.3.2 Maxwell's Third relation

Taking the total differential, we get

$$dF = dU - T\,dS - S\,dT. \tag{7.25}$$

Using the fundamental relation $dU = T\,dS - P\,dV$, we obtain

$$dF = -P\,dV - S\,dT. \tag{7.26}$$

we see that the natural variables of Helmholtz free energy are

$$F = F(T, V). \tag{7.27}$$

Consequently,

$$S = -\left(\frac{\partial F}{\partial T}\right)_V, \qquad P = -\left(\frac{\partial F}{\partial V}\right)_T. \tag{7.28}$$

Since $F(T, V)$ is a state function, its mixed second derivatives are equal:

$$\frac{\partial}{\partial V}\left(\frac{\partial F}{\partial T}\right)_V = \frac{\partial}{\partial T}\left(\frac{\partial F}{\partial V}\right)_T. \tag{7.29}$$

Substituting the thermodynamic derivatives, we obtain the Maxwell relation:

$$\boxed{\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V} \tag{7.30}$$

This is the Maxwell relation associated with Helmholtz free energy.

7.4 Gibbs Free Energy G

Gibbs free energy is the thermodynamic potential most useful for systems in contact with a heat and pressure reservoir, i.e., at constant temperature T and constant pressure P.

In such situations, the equilibrium state corresponds to minimum Gibbs free energy.

Gibbs Free Energy

Physically, G represents the maximum useful (non–$P\,dV$) work that a system can perform when the temperature and pressure are held constant.

7.4.1 Definition of Gibbs Free Energy

Gibbs free energy is defined as

$$\boxed{G = U + PV - TS} \tag{7.31}$$

Equivalently, using $H = U + PV$,

$$G = H - TS \tag{7.32}$$

7.4.2 Maxwell's Fourth relation

Taking the total differential, we have

$$dG = dU + P\,dV + V\,dP - T\,dS - S\,dT. \tag{7.33}$$

Using the fundamental relation $dU = T\,dS - P\,dV$, we simplify:

$$dG = -S\,dT + V\,dP \tag{7.34}$$

From the differential, it is clear that the natural variables of Gibbs free energy are

$$G = G(T, P) \tag{7.35}$$

Correspondingly, the thermodynamic derivatives are

$$S = -\left(\frac{\partial G}{\partial T}\right)_P, \qquad V = \left(\frac{\partial G}{\partial P}\right)_T \tag{7.36}$$

Since $G(T, P)$ is a state function, its mixed second derivatives are equal:

$$\frac{\partial}{\partial P}\left(\frac{\partial G}{\partial T}\right)_P = \frac{\partial}{\partial T}\left(\frac{\partial G}{\partial P}\right)_T \tag{7.37}$$

Substituting the thermodynamic derivatives, we obtain the Maxwell relation:

$$\boxed{\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P} \tag{7.38}$$

This is the Maxwell relation associated with Gibbs free energy.

Chapter 8

Thermodynamic relations

8.1 $C_P - C_V$ Relations

$$C_V = \left(\frac{\delta Q}{\partial T}\right)_V = T\left(\frac{\partial S}{\partial T}\right)_V \tag{8.1}$$

$$C_P = \left(\frac{\delta Q}{\partial T}\right)_P = T\left(\frac{\partial S}{\partial T}\right)_P \tag{8.2}$$

Taking Entropy S: as a function of T and V

$$S = S(T, V) \tag{8.3}$$

$$dS = \left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV \tag{8.4}$$

Which implies:

$$\left(\frac{\partial S}{\partial T}\right)_P = \left(\frac{\partial S}{\partial T}\right)_V + \left(\frac{\partial S}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P \tag{8.5}$$

from eqn 8.2 and 8.5:

$$C_P = \left(\frac{\delta Q}{\partial T}\right)_P = T\left(\frac{\partial S}{\partial T}\right)_V + T\left(\frac{\partial S}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P \tag{8.6}$$

$$C_P = \left(\frac{\delta Q}{\partial T}\right)_P = C_V + T\left(\frac{\partial S}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P \tag{8.7}$$

Using maxwell's relation:

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V \tag{8.8}$$

$$\boxed{C_P - C_V = T\left(\frac{\partial P}{\partial T}\right)_V \left(\frac{\partial V}{\partial T}\right)_P} \tag{8.9}$$

8.2 The $T\,dS$ Equations

Entropy is a state function and may be expressed in terms of different pairs of independent thermodynamic variables. Depending on the choice of variables, different forms of the $T\,dS$ equation are obtained. We derive all three standard forms below.

8.2.1 Entropy as a Function of T and V

Let the entropy be expressed as

$$S = S(T, V). \tag{8.10}$$

The total differential of entropy is

$$dS = \left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV. \tag{8.11}$$

Multiplying throughout by T,

$$T\,dS = T\left(\frac{\partial S}{\partial T}\right)_V dT + T\left(\frac{\partial S}{\partial V}\right)_T dV. \tag{8.12}$$

From the definition of heat capacity at constant volume,

$$C_V = T\left(\frac{\partial S}{\partial T}\right)_V, \tag{8.13}$$

and using the Maxwell relation

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V, \tag{8.14}$$

we obtain

$$\boxed{T\,dS = C_V\,dT + T\left(\frac{\partial P}{\partial T}\right)_V dV.} \tag{8.15}$$

This is the first $T\,dS$ equation.

8.2.2 Entropy as a Function of T and P

Let entropy be written as

$$S = S(T, P). \tag{8.16}$$

The total differential is

$$dS = \left(\frac{\partial S}{\partial T}\right)_P dT + \left(\frac{\partial S}{\partial P}\right)_T dP. \tag{8.17}$$

Multiplying by T,

$$T\,dS = T\left(\frac{\partial S}{\partial T}\right)_P dT + T\left(\frac{\partial S}{\partial P}\right)_T dP. \tag{8.18}$$

Using the definition of heat capacity at constant pressure,

$$C_P = T\left(\frac{\partial S}{\partial T}\right)_P, \tag{8.19}$$

and the Maxwell relation

$$\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P, \tag{8.20}$$

we obtain

$$\boxed{T\,dS = C_P\,dT - T\left(\frac{\partial V}{\partial T}\right)_P dP.} \tag{8.21}$$

This is the second $T\,dS$ equation.

8.2.3 Entropy as a Function of P and V

Let entropy be expressed as

$$S = S(P, V). \tag{8.22}$$

The total differential is

$$dS = \left(\frac{\partial S}{\partial P}\right)_V dP + \left(\frac{\partial S}{\partial V}\right)_P dV. \tag{8.23}$$

Multiplying by T,

$$T\,dS = T\left(\frac{\partial S}{\partial P}\right)_V dP + T\left(\frac{\partial S}{\partial V}\right)_P dV. \tag{8.24}$$

Using thermodynamic identities,

$$T\left(\frac{\partial S}{\partial P}\right)_V = C_V\left(\frac{\partial T}{\partial P}\right)_V, \tag{8.25}$$

$$T\left(\frac{\partial S}{\partial V}\right)_P = C_P\left(\frac{\partial T}{\partial V}\right)_P, \tag{8.26}$$

we obtain

$$\boxed{T\,dS = C_V\left(\frac{\partial T}{\partial P}\right)_V dP + C_P\left(\frac{\partial T}{\partial V}\right)_P dV.} \tag{8.27}$$

This is the third $T\,dS$ equation.

8.3 Clapeyron Equation

The Clapeyron equation describes the slope of a phase boundary in the pressure–temperature (P–T) diagram. It is derived from the condition of thermodynamic equilibrium between two coexisting phases.

Thermodynamic Condition for Phase Equilibrium

Consider a system in which two phases, denoted by α and β, coexist in equilibrium. The necessary and sufficient condition for phase equilibrium is the equality of chemical potentials:

$$\mu^{(\alpha)}(T, P) = \mu^{(\beta)}(T, P). \tag{8.28}$$

Along the phase coexistence curve, this equality must hold for infinitesimal changes in T and P. Hence,

$$d\mu^{(\alpha)} = d\mu^{(\beta)}. \tag{8.29}$$

Differential of Chemical Potential

For a simple compressible system, the differential of chemical potential is given by Gibbs Dulhem Equation:

$$d\mu = -S_m\, dT + V_m\, dP, \tag{8.30}$$

where S_m and V_m are the molar entropy and molar volume, respectively.

Applying this to the two phases,

$$d\mu^{(\alpha)} = -S_m^{(\alpha)}\, dT + V_m^{(\alpha)}\, dP, \tag{8.31}$$

$$d\mu^{(\beta)} = -S_m^{(\beta)}\, dT + V_m^{(\beta)}\, dP. \tag{8.32}$$

Since $d\mu^{(\alpha)} = d\mu^{(\beta)}$, we obtain

$$\left[S_m^{(\beta)} - S_m^{(\alpha)}\right] dT = \left[V_m^{(\beta)} - V_m^{(\alpha)}\right] dP. \tag{8.33}$$

Clapeyron Equation

Rearranging the above relation,

$$\boxed{\frac{dP}{dT} = \frac{S_m^{(\beta)} - S_m^{(\alpha)}}{V_m^{(\beta)} - V_m^{(\alpha)}}} \tag{8.34}$$

Using the definition of latent heat per mole,

$$L = T\left(S_m^{(\beta)} - S_m^{(\alpha)}\right), \tag{8.35}$$

the Clapeyron equation can be written in the standard form

$$\boxed{\frac{dP}{dT} = \frac{L}{T\left(V_m^{(\beta)} - V_m^{(\alpha)}\right)}} \tag{8.36}$$

This is the *Clapeyron equation.* It is valid for all first-order phase transitions such as solid–liquid, liquid–vapour, and solid–vapour equilibrium.

8.3.1 Clausius–Clapeyron Equation (Liquid–Vapour Transition)

For a liquid–vapour phase transition,

$$V_m^{(v)} \gg V_m^{(l)}. \tag{8.37}$$

Hence,

$$V_m^{(v)} - V_m^{(l)} \approx V_m^{(v)}. \tag{8.38}$$

Assuming the vapour behaves as an ideal gas,

$$V_m^{(v)} = \frac{RT}{P}. \tag{8.39}$$

Substituting into the Clapeyron equation,

$$\frac{dP}{dT} = \frac{L}{T} \cdot \frac{P}{RT} \tag{8.40}$$

$$= \frac{LP}{RT^2}. \tag{8.41}$$

Rearranging,

$$\boxed{\frac{dP}{P} = \frac{L}{RT^2}\, dT} \tag{8.42}$$

This is the *Clausius–Clapeyron equation* in differential form.

Integrated Clausius–Clapeyron Equation

If the latent heat L is assumed to be temperature independent, integration yields

$$\int_{P_1}^{P_2} \frac{dP}{P} = \frac{L}{R} \int_{T_1}^{T_2} \frac{dT}{T^2}. \tag{8.43}$$

Performing the integrations,

$$\boxed{\ln\left(\frac{P_2}{P_1}\right) = -\frac{L}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)} \tag{8.44}$$

This form is widely used to determine latent heat from vapour pressure data.

8.4 Joule–Thomson Expansion

The Joule–Thomson effect refers to the change in temperature of a real gas when it is forced to expand from a region of high pressure to a region of low pressure through a porous plug or throttle valve under adiabatic conditions. Since the system is thermally insulated, no heat is exchanged with the surroundings during the process.

Such an expansion is commonly known as *throttling expansion*. The Joule–Thomson effect is of fundamental importance in refrigeration, liquefaction of gases, and cryogenic applications.

For most real gases at ordinary temperatures, the gas cools upon expansion. However, gases such as hydrogen and helium exhibit heating under similar conditions unless they are cooled below a characteristic temperature called the *inversion temperature*.

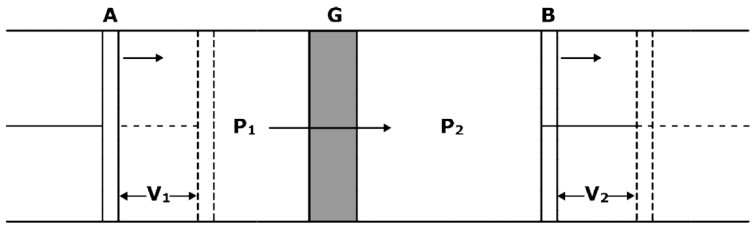


Figure 8.1: The porous experiment

8.4.1 Isoenthalpic Nature of Joule–Thomson Expansion

Consider a gas flowing steadily through a porous plug from an initial state (P_1, V_1, T_1) to a final state (P_2, V_2, T_2) in an insulated tube.

The work done on the gas at the inlet is

$$W_1 = P_1 V_1, \tag{8.45}$$

and the work done by the gas at the outlet is

$$W_2 = P_2 V_2. \tag{8.46}$$

Since the process is adiabatic and no macroscopic kinetic energy change is involved, the first law of thermodynamics gives

$$U_2 - U_1 = P_1 V_1 - P_2 V_2. \tag{8.47}$$

Rearranging,

$$U_1 + P_1 V_1 = U_2 + P_2 V_2. \tag{8.48}$$

Using the definition of enthalpy $H = U + PV$, we obtain

$$H_1 = H_2, \tag{8.49}$$

or

$$\boxed{\Delta H = 0.} \tag{8.50}$$

Thus, Joule–Thomson expansion is an *isoenthalpic* process.

8.4.2 Joule–Thomson Coefficient

The *Joule–Thomson coefficient* μ_{JT} is defined as

$$\boxed{\mu_{JT} = \left(\frac{\partial T}{\partial P}\right)_H} \tag{8.51}$$

The quantity μ_{JT} is called the *Joule–Thomson coefficient*. It represents the rate of change of temperature with pressure during an isoenthalpic process.

- $\mu_{JT} > 0$: cooling on expansion
- $\mu_{JT} < 0$: heating on expansion

- $\mu_{JT} = 0$: no temperature change

Starting from the thermodynamic identity,

$$dH = T\,dS + V\,dP, \tag{8.52}$$

Since we are interested in change in temperature with pressure, we write entropy as

$$S = S(T, P)$$

$$dS = \left(\frac{\partial S}{\partial T}\right)_P dT + \left(\frac{\partial S}{\partial P}\right)_T dP.$$

Substituting,

$$dH = T\left(\frac{\partial S}{\partial T}\right)_P dT + \left[V + T\left(\frac{\partial S}{\partial P}\right)_T\right] dP. \tag{8.53}$$

Using the Maxwell relation,

$$\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P, \tag{8.54}$$

and Heat capacity at constant Pressure:

$$C_P = \left(\frac{\delta Q}{\delta T}\right)_P = T\left(\frac{\partial S}{\partial T}\right)_P \tag{8.55}$$

we get

$$dH = C_P dT + \left[V - T\left(\frac{\partial V}{\partial T}\right)_P\right] dP. \tag{8.56}$$

For Joule-Thomson Expansion, $dH = 0$, implies:

$$C_P dT + \left[V - T\left(\frac{\partial V}{\partial T}\right)_P\right] dP = 0 \tag{8.57}$$

the Joule–Thomson coefficient may be written as

$$\boxed{\mu_{JT} \equiv \left(\frac{\partial T}{\partial P}\right)_H = -\frac{1}{C_P}\left(\frac{\partial H}{\partial P}\right)_T = \frac{1}{C_P}\left[T\left(\frac{\partial V}{\partial T}\right)_P - V\right].} \tag{8.58}$$

8.4.3 Ideal Gas Case

For an ideal gas,

$$V = \frac{nRT}{P}, \qquad \left(\frac{\partial V}{\partial T}\right)_P = \frac{nR}{P}.$$

Substituting,

$$\mu_{JT} = \frac{1}{C_P}\left(\frac{nRT}{P} - \frac{nRT}{P}\right) = 0.$$

8.4.4 Joule–Thomson Coefficient for van der Waals

For one mole of a van der Waals gas,

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT. \tag{8.59}$$

For low pressures, neglecting higher-order terms,

$$V \approx \frac{RT}{P} + b - \frac{a}{RT}. \tag{8.60}$$

Differentiating with respect to T at constant P,

$$\left(\frac{\partial V}{\partial T}\right)_P = \frac{R}{P} + \frac{a}{RT^2}. \tag{8.61}$$

Substituting into the expression for μ_{JT},

$$\mu_{JT} = \frac{1}{C_P}\left(\frac{2a}{RT} - b\right). \tag{8.62}$$

Inversion Temperature

The inversion temperature T_i is defined as the temperature at which the Joule–Thomson coefficient changes sign, i.e.,

$$\mu_{JT} = 0. \tag{8.63}$$

From the above expression,

$$\frac{2a}{RT_i} = b, \tag{8.64}$$

which gives

$$\boxed{T_i = \frac{2a}{Rb}.} \tag{8.65}$$

For $T < T_i$, the gas cools on expansion ($\mu_{JT} > 0$), while for $T > T_i$, the gas heats up ($\mu_{JT} < 0$).

8.5 Cooling by Adiabatic Demagnetization

A paramagnetic salt contains ions with magnetic moments which, in the presence of an external magnetic field, tend to align with the field. At ordinary temperatures, thermal agitation keeps these moments largely disordered, resulting in high magnetic entropy.

When a strong magnetic field is applied *isothermally*, the magnetic moments align partially with the field and the magnetic entropy decreases. If the system is then thermally isolated and the magnetic field is reduced *adiabatically*, the entropy must remain constant. As a result, thermal disorder decreases and the temperature of the salt falls. This phenomenon is known as *adiabatic demagnetization*.

8.5.1 Thermodynamics of magnetic system

For a magnetic system, the first law of thermodynamics may be written as (Just $-PdV$ will be replaced by $+HdM$)

$$dU = TdS + H\,dM, \tag{8.66}$$

where H is the magnetic field and M is the magnetization.

Using entropy as a function of temperature T and magnetic field H,

$$dS = \left(\frac{\partial S}{\partial T}\right)_H dT + \left(\frac{\partial S}{\partial H}\right)_T dH. \tag{8.67}$$

For an adiabatic process,

$$dS = 0 \tag{8.68}$$

Hence,

$$\left(\frac{\partial S}{\partial T}\right)_H dT = -\left(\frac{\partial S}{\partial H}\right)_T dH.$$

8.5.2 Magnetic Entropy Relations

Using Maxwell's thermodynamic relation for magnetic systems,

$$\left(\frac{\partial S}{\partial H}\right)_T = \left(\frac{\partial M}{\partial T}\right)_H.$$

Also,

$$\left(\frac{\partial S}{\partial T}\right)_H = \frac{C_H}{T},$$

where C_H is the heat capacity at constant magnetic field.

Substituting,

$$\frac{C_H}{T}\,dT = -\left(\frac{\partial M}{\partial T}\right)_H dH.$$

8.5.3 Expression for Cooling

For a paramagnetic salt obeying Curie's law,

$$M = \frac{CH}{T},$$

where C is Curie's constant.

Therefore,

$$\left(\frac{\partial M}{\partial T}\right)_H = -\frac{CH}{T^2}.$$

Substituting into the adiabatic condition,

$$\frac{C_H}{T}\,dT = \frac{CH}{T^2}dH.$$

Rearranging,

$$\frac{dT}{T} = \frac{C}{C_H}\frac{H\,dH}{T^2}.$$

Integrating from T_i, H_i to T_f, H_f,

$$T_f^2 = T_i^2 + \frac{C}{C_H}(H_f^2 - H_i^2) \tag{8.69}$$

For sufficiently low temperatures, C_H is approximately constant, and integration gives

$$\boxed{\frac{T}{H} = \text{constant}}$$

or equivalently,

$$\boxed{T_f = T_i \frac{H_f}{H_i}}.$$

Thus, during adiabatic demagnetization, the temperature of a paramagnetic salt is directly proportional to the applied magnetic field. By reducing the magnetic field from H_i to H_f, extremely low temperatures (of the order of millikelvin) can be achieved.

This method is widely used to attain temperatures well below those obtainable by conventional refrigeration techniques.

Chapter 9

Third Law of Thermodynamics

The first and second laws of thermodynamics introduce the concepts of energy conservation and entropy, but they do not fix the *absolute value* of entropy. The third law provides this missing reference by specifying the behavior of entropy as the temperature approaches absolute zero.

This law plays a crucial role in low-temperature physics, chemical thermodynamics, and the definition of absolute entropy.

From the second law, entropy is defined through

$$dS = \frac{\delta Q_{\text{rev}}}{T}. \tag{9.1}$$

This definition determines only *changes* in entropy. To assign an absolute value of entropy, a reference state is required. The third law supplies this reference by specifying the entropy at absolute zero.

9.1 Planck's Statement of the Third Law

Third Law of Thermodynamics (Planck Statement)

The entropy of a perfect crystalline substance approaches zero as the temperature approaches absolute zero:

$$\boxed{\lim_{T\to 0} S = 0} \tag{9.2}$$

A *perfect crystal* is one with a unique ground state (no residual disorder).

9.1.1 Mathematical Formulation

At constant volume, the reversible heat supplied to a system is

$$\delta Q_{\text{rev}} = C_V(T)\, dT, \tag{9.3}$$

where C_V is the heat capacity at constant volume.

Hence,

$$dS = \frac{C_V(T)}{T}\, dT. \tag{9.4}$$

Integrating from absolute zero to temperature T,

$$\boxed{S(T) - S(0) = \int_0^T \frac{C_V(T')}{T'}\, dT'} \tag{9.5}$$

This equation is the central mathematical expression underlying the third law.

9.1.2 Vanishing of Heat Capacity at Absolute Zero

For the entropy at $T = 0$ to be finite, the integral in Eq. (9.5) must converge. This requires

$$\boxed{\lim_{T\to 0} C_V(T) = 0} \tag{9.6}$$

Thus, the third law implies that the heat capacity of all systems must vanish as the temperature approaches absolute zero.

Physical Meaning

At very low temperatures, thermal motion freezes out and the system cannot absorb energy efficiently, leading to vanishing heat capacity.

9.1.3 Zero Entropy of a Perfect Crystal

From statistical mechanics, entropy is given by Boltzmann's formula:

$$S = k_B \ln W, \tag{9.7}$$

where W is the number of accessible microstates.

For a perfect crystal at $T = 0$,

$$W = 1, \tag{9.8}$$

since the ground state is unique.

Hence,

$$\boxed{S(0) = 0} \tag{9.9}$$

This provides a microscopic justification of the third law.

9.2 Nernst Heat Theorem

An alternative formulation of the third law is the *Nernst heat theorem*.

Nernst Heat Theorem

The entropy change associated with any isothermal reversible process tends to zero as the temperature approaches absolute zero:

$$\boxed{\lim_{T\to 0} \Delta S = 0} \tag{9.10}$$

Mathematical Justification

For an isothermal reversible process,

$$\Delta S = \int \frac{\delta Q_{\text{rev}}}{T}. \tag{9.11}$$

As $T \to 0$, the heat exchanged $\delta Q_{\text{rev}} \to 0$ due to vanishing heat capacity, hence $\Delta S \to 0$.

9.2.1 Unattainability of Absolute Zero

Another equivalent statement of the third law is the *unattainability principle*.

Unattainability Principle

Absolute zero cannot be reached in a finite number of thermodynamic processes.

Mathematical Argument

A temperature change is given by

$$dT = \frac{\delta Q}{C(T)}. \tag{9.12}$$

As $T \to 0$, $C(T) \to 0$, so achieving a finite decrease in temperature requires an infinite number of steps.

9.3 Residual Entropy

If a system has a degenerate ground state ($W > 1$), then

$$S(0) = k_B \ln W \neq 0. \tag{9.13}$$

This is known as *residual entropy* and occurs in systems with frozen disorder, such as glasses.

Important Note

The third law strictly applies to *perfect crystals*. Residual entropy does not violate the law but reflects incomplete ordering.

The third law thus completes the logical structure of thermodynamics and provides the foundation for low-temperature physics.

Chapter 10

Calorimetry and Thermometry

10.1 Fourier's Law of Heat Conduction

The rate of heat flow through a material is directly proportional to the cross-sectional area and the temperature gradient across the material, provided the temperature difference is small and steady-state conditions exist.

Mathematically,

$$\frac{dQ}{dt} = -kA\frac{dT}{dx} \tag{10.1}$$

where,

- $\frac{dQ}{dt}$ = rate of heat transfer (W)
- k = thermal conductivity of the material
- A = cross-sectional area
- $\frac{dT}{dx}$ = temperature gradient

The negative sign indicates that heat flows in the direction of decreasing temperature.

Vector Form of Fourier's Law

In three dimensions, Fourier's law can be written in vector form as:

$$\vec{q} = -k\nabla T \tag{10.2}$$

where $\vec{q}$ represents heat flux (heat flow per unit area per unit time).

Thermal Conductivity

The amount of heat transmitted per unit time through a unit area of a material when the temperature gradient is unity.

The SI unit of thermal conductivity is:

$$\mathrm{W\ m^{-1}K^{-1}} \tag{10.3}$$

10.2 Specific Heat

The quantity of heat required to raise the temperature of a body is directly proportional to its mass and the change in temperature. Thus,

$$Q \propto m\Delta T \tag{10.4}$$

Introducing a proportionality constant c, we obtain:

$$Q = mc\Delta T \tag{10.5}$$

where,

- Q = heat supplied
- m = mass of the substance
- c = specific heat capacity
- ΔT = change in temperature

This equation is fundamental in calorimetry and thermal measurements.

Water Equivalent

Water equivalent is a useful concept in calorimetry. It is defined as the mass of water that would absorb the same amount of heat as the given body for the same temperature change.

If a body has heat capacity C, its water equivalent W is:

$$W = \frac{C}{c_w} \tag{10.6}$$

where c_w is the specific heat of water.

10.3 Determination of Thermal Conductivity of Good Conductors: Searle's Method

Searle's method is commonly used to determine the thermal conductivity of good conductors such as copper, brass, and aluminum.

10.3.1 Principle

When a uniform metal rod is heated at one end and cooled at the other end, heat flows along the rod from the hot end to the cold end.

10.3.2 Theory and Derivation

According to Fourier's law, the rate of heat conduction along the rod is:

$$\frac{dQ}{dt} = kA\frac{T_1 - T_2}{x} \tag{10.7}$$

The heat gained by water per second is:

$$\frac{dQ}{dt} = mc(\theta_2 - \theta_1) \tag{10.8}$$

Under steady-state conditions,

$$kA\frac{T_1 - T_2}{x} = mc(\theta_2 - \theta_1) \tag{10.9}$$

Therefore, thermal conductivity is given by:

$$k = \frac{mc(\theta_2 - \theta_1)x}{A(T_1 - T_2)} \tag{10.10}$$

10.4 Determination of Thermal Conductivity of Bad Conductors: Lee's Disc Method

Theory and Derivation

According to Fourier's law, the heat conducted per second through the bad conductor at steady state is:

$$Q = kA\frac{T_1 - T_2}{x} \tag{10.11}$$

The rate of heat loss from the disc at temperature T_2 is:

$$Q = mc\left(\frac{dT}{dt}\right)_{T=T_2} \tag{10.12}$$

At steady state,

$$kA\frac{T_1 - T_2}{x} = mc\left(\frac{dT}{dt}\right)_{T=T_2} \tag{10.13}$$

Therefore, thermal conductivity is:

$$k = \frac{mcx}{A(T_1 - T_2)}\left(\frac{dT}{dt}\right)_{T=T_2} \tag{10.14}$$

10.5 Newton's Law of Cooling

The rate of heat loss is proportional to temperature difference:

$$\frac{dQ}{dt} \propto (T - T_s) \tag{10.15}$$

Since heat lost changes the temperature of the body,

$$dQ = mc\,dT \tag{10.16}$$

Therefore,

$$mc\frac{dT}{dt} = -k(T - T_s) \tag{10.17}$$

or,

$$\frac{dT}{dt} = -K(T - T_s) \tag{10.18}$$

Temperature–Time Relation

Rearranging,

$$\frac{dT}{T - T_s} = -K\,dt \tag{10.19}$$

Integrating,

$$\int \frac{dT}{T - T_s} = -K \int dt \tag{10.20}$$

$$\ln(T - T_s) = -Kt + C \tag{10.21}$$

At time $t = 0$, temperature is T_0, therefore,

$$C = \ln(T_0 - T_s) \tag{10.22}$$

Thus,

$$\ln(T - T_s) = \ln(T_0 - T_s) - Kt \tag{10.23}$$

or,

$$T - T_s = (T_0 - T_s)e^{-Kt} \tag{10.24}$$

10.6 Thermometers

Liquid-in-Glass Thermometers

The operation of these thermometers is based on the **differential thermal expansion** of a liquid relative to its glass envelope. When heat is applied, the volume of the liquid increases following the relation:

$$V_t = V_0(1 + \gamma\Delta T) \tag{10.25}$$

Because the volumetric expansion coefficient of the liquid (γ_l) is much larger than that of the glass (γ_g), the liquid is displaced into the capillary. For a uniform bore of cross-sectional area A, the height of the column h relates to temperature T as:

$$h = \frac{V_0(\gamma_l - \gamma_g)}{A}(T - T_0) \tag{10.26}$$

This linear relationship allows for the graduation of the thermometer stem into equal temperature intervals.

Gas Thermometry

Gas thermometers are based on the properties of an ideal gas.

$$PV = nRT \tag{10.27}$$

In a **Constant Volume Gas Thermometer**, the volume is fixed, and the temperature is determined by measuring the pressure:

$$T = T_{ref}\left(\frac{P}{P_{ref}}\right) \tag{10.28}$$

10.7 Ohm's Law and Electrical Conduction

Mathematically,

$$V \propto I \tag{10.29}$$

$$V = IR \tag{10.30}$$

or in Vector Form, Current density $\mathbf{j}$ due to Applied Electric Field $\mathbf{E}$

$$\mathbf{j} = \sigma\mathbf{E} \tag{10.31}$$

Where σ is electrical conductivity.

10.7.1 Joule Heating and Energy Dissipation

Starting from Ohm's Law:

$$V = IR \tag{10.32}$$

The electrical power dissipated:

$$P = VI \tag{10.33}$$

Substituting:

$$P = I^2R \tag{10.34}$$

Total heat:

$$Q = \int_0^t I^2R\,dt \tag{10.35}$$

Joule heating:

$$Q = I^2Rt \tag{10.36}$$

10.7.2 Platinum Resistance Thermometer (PRT)

For small temperature ranges:

$$R_T = R_0(1 + \alpha T) \tag{10.37}$$

Temperature Coefficient of Resistance

$$\alpha = \frac{R_{100} - R_0}{100 R_0} \tag{10.38}$$

For standard platinum thermometers:

$$\alpha \approx 0.00385\,^\circ C^{-1} \tag{10.39}$$

Sensitivity

$$S = \frac{dR}{dT} \tag{10.40}$$

10.7.3 Thermistors

Electrical conductivity:

$$\sigma = ne\mu \tag{10.41}$$

Carrier concentration:

$$n \propto e^{-E_g/2kT} \tag{10.42}$$

Arrhenius dependence:

$$\sigma = \sigma_0 e^{-E_a/kT} \tag{10.43}$$

Resistance relation:

$$R = R_0 e^{E_a/kT} \tag{10.44}$$

Practical thermistor relation:

$$R = R_0\, e^{B\left(\frac{1}{T} - \frac{1}{T_0}\right)} \tag{10.45}$$

Temperature Coefficient of Resistance

$$\alpha = \frac{1}{R}\frac{dR}{dT} \tag{10.46}$$

$$\alpha = -\frac{B}{T^2} \tag{10.47}$$

Sensitivity of Thermistors

$$S = \frac{dR}{dT} \tag{10.48}$$

$$\frac{dR}{dT} = -\frac{B}{T^2}R \tag{10.49}$$

Steinhart–Hart Equation

$$\frac{1}{T} = A + B\ln R + C(\ln R)^3 \tag{10.50}$$

10.8 Seebeck Effect

The Seebeck effect is a thermoelectric phenomenon in which an electromotive force (emf) is generated in a closed circuit made of two dissimilar conductors when their junctions are maintained at different temperatures. This effect forms the basis of thermocouples used for temperature measurement.

Experimental Observation

Consider two different metallic conductors A and B joined to form two junctions. One junction is kept at temperature T_1 (hot junction) and the other at temperature T_2 (cold junction).

When the two junctions are maintained at different temperatures, an emf is generated in the circuit, causing a current to flow if the circuit is closed.

Seebeck Coefficient

The magnitude of the thermoelectric emf depends on the temperature difference between the junctions. For small temperature differences, the emf is approximately proportional to the temperature difference:

$$E \propto (T_1 - T_2)$$

Introducing a proportionality constant S, called the Seebeck coefficient or thermoelectric power:

$$E = S(T_1 - T_2) \tag{10.51}$$

The Seebeck coefficient depends on the nature of the materials used.

Differential Form

For a small temperature difference dT, the emf generated is:

$$dE = S\,dT \tag{10.52}$$

Thus,

$$S = \frac{dE}{dT} \tag{10.53}$$

The unit of Seebeck coefficient is volt per kelvin (V/K).

Microscopic Explanation

The Seebeck effect arises due to the diffusion of charge carriers (electrons or holes) from the hot region to the cold region. At higher temperatures, charge carriers possess higher kinetic energy and diffuse toward the cooler region, creating a potential difference.

Approximate Quadratic Relation

For small temperature ranges, the emf can be expressed as:

$$E = aT + \frac{1}{2}bT^2 \tag{10.54}$$

where

- a = first thermoelectric coefficient
- b = second thermoelectric coefficient
- T = temperature difference between junctions

This empirical relation is widely used in thermocouple calibration.

Microscopic Theory

From electron transport theory, the Seebeck coefficient for metals can be written approximately as:

$$S = \frac{\pi^2 k^2 T}{3e} \left(\frac{1}{n} \frac{dn}{dE} \right)_{E_F} \tag{10.55}$$

where

- k = Boltzmann constant
- e = electronic charge
- n = carrier concentration
- E_F = Fermi energy

This expression shows that thermoelectric effects depend on the energy dependence of charge carrier density near the Fermi level.

10.8.1 Thermocouple emf Between Two Metals

Different materials have different electron densities and energy distributions, resulting in net emf when two materials are joined.

Consider two different metals A and B forming a closed circuit with junctions at temperatures T_1 and T_2.

The net emf generated is:

$$E_{AB} = \int_{T_1}^{T_2} (S_A - S_B)\, dT \tag{10.56}$$

Thus the thermocouple emf depends on the difference of Seebeck coefficients of the two materials.

10.9 Peltier Effect

The Peltier effect is the reverse of the Seebeck effect and describes the absorption or evolution of heat at the junction of two dissimilar conductors when an electric current passes through the circuit.

When a current flows across a junction formed by two different materials, heat may either be absorbed (cooling) or released (heating), depending on the direction of current. This phenomenon is known as the Peltier effect.

Different materials possess different charge carrier energy levels. When charge carriers move from one material to another, they must either gain or lose energy to adapt to the new environment. This energy exchange occurs in the form of heat at the junction.

Thus:

- Heat is absorbed at one junction (cooling junction)
- Heat is released at the other junction (heating junction)

Peltier Coefficient

The heat absorbed or evolved at a junction is proportional to the electric current flowing through it.

$$Q = \Pi I t \tag{10.57}$$

where:

- Q = heat absorbed or released
- Π = Peltier coefficient
- I = electric current
- t = time

It is important to distinguish between Peltier heating and Joule heating:

- Joule heating occurs throughout a conductor and is proportional to I^2R
- Peltier heating occurs only at junctions and is proportional to I

The Peltier effect provides the foundation for thermoelectric refrigeration and plays a crucial role in the unified understanding of heat flow and electric current transport in materials.

10.10 Unified Theory

Ohm's Law $\mathbf{J} = \sigma \mathbf{E}$ $\mathbf{J}$: Electric current density σ : Electrical conductivity $\mathbf{E}$: Electric field	**Seebeck Effect** $\mathbf{E} = -S\nabla T$ S : Seebeck coefficient ∇T : Temperature gradient
Peltier Effect $\mathbf{q} = \Pi \mathbf{J}$ $\mathbf{q}$: Heat current density Π : Peltier coefficient $\mathbf{J}$: Electric current density	**Fourier's Law** $\mathbf{q} = -k\nabla T$ $\mathbf{q}$: Heat flux k : Thermal conductivity ∇T : Temperature gradient

Matrix Formulation of Thermoelectric Transport

In thermoelectric materials, electric current and heat flow are coupled processes. The generalized transport equations can be written in matrix form as:

$$\begin{bmatrix} \mathbf{J} \\ \mathbf{q} \end{bmatrix} = \begin{bmatrix} \sigma & \sigma S \\ \Pi\sigma & k + \Pi\sigma S \end{bmatrix} \begin{bmatrix} \mathbf{E} \\ -\nabla T \end{bmatrix} \tag{10.58}$$

where

- $\mathbf{J}$ = electric current density
- $\mathbf{q}$ = heat flux

Thus thermoelectricity is a coupled transport phenomenon.

10.10.1 Onsager Reciprocity Relation

Thermodynamic symmetry requires that the cross-coefficients satisfy:

$$\Pi = ST \tag{10.59}$$

This important relation is known as the Kelvin–Onsager relation.

Chapter 11

Black Body Radiation

11.1 Basic Definitions

Before studying black body radiation, we introduce some important radiative quantities.

Total Energy Density

The total energy density u is defined as the total radiant energy per unit volume inside an enclosure:

$$u = \frac{\text{Total radiant energy}}{\text{Volume}} \tag{11.1}$$

Spectral Energy Density

The spectral energy density u_λ is the energy per unit volume per unit wavelength range:

$$u_\lambda = \frac{du}{d\lambda} \tag{11.2}$$

such that

$$u = \int_0^\infty u_\lambda \, d\lambda \tag{11.3}$$

Emissive Power

The emissive power E is defined as the total radiant energy emitted per unit area per unit time:

$$E = \frac{\text{Energy emitted}}{\text{Area} \times \text{time}} \tag{11.4}$$

Spectral Emissive Power

The spectral emissive power E_λ is the energy emitted per unit area per unit time per unit wavelength:

$$E_\lambda = \frac{dE}{d\lambda} \tag{11.5}$$

and

$$E = \int_0^\infty E_\lambda \, d\lambda \tag{11.6}$$

Emissivity

Emissivity e of a surface is defined as the ratio of emissive power of the surface to that of a perfect black body at the same temperature:

$$e = \frac{E}{E_b} \tag{11.7}$$

Emissivity of a Black Body

For a perfect black body,

$$e = 1 \tag{11.8}$$

A black body is thus defined as an ideal system that absorbs all incident radiation and emits the maximum possible radiation at a given temperature.

Spectral Absorptivity

Spectral absorptivity a_λ is defined as the fraction of incident radiation of wavelength λ that is absorbed:

$$a_\lambda = \frac{\text{Energy absorbed at } \lambda}{\text{Energy incident at } \lambda} \tag{11.9}$$

Reflectivity

Reflectivity r is the fraction of incident radiation that is reflected:

$$r = \frac{\text{Reflected energy}}{\text{Incident energy}} \tag{11.10}$$

Transmissivity

Transmissivity t is the fraction of incident radiation that passes through the material:

$$t = \frac{\text{Transmitted energy}}{\text{Incident energy}} \tag{11.11}$$

11.2 Conservation of Radiant Energy

When radiation is incident on a surface, it can be partially absorbed, reflected, or transmitted. By conservation of energy, the sum of all fractions must be unity:

$$r + a + t = 1 \tag{11.12}$$

For an opaque body, transmission is zero ($t = 0$), hence:

$$r + a = 1 \tag{11.13}$$

For a perfect black body:

$$a = 1, \quad r = 0, \quad t = 0 \tag{11.14}$$

Thus, a black body absorbs all incident radiation and reflects or transmits none.

11.3 Kirchhoff's Law of Thermal Radiation

Gustav Kirchoff

Gustav Kirchhoff was a German physicist who made fundamental contributions to thermal radiation and spectroscopy. In 1859, he formulated Kirchhoff's law of thermal radiation, which states that for a body in thermal equilibrium, the ratio of emissive power to absorptive power is the same for all bodies and depends only on temperature and wavelength.
This insight led to the concept of an ideal emitter known as a *black body*, which absorbs and emits radiation perfectly.
Kirchhoff's work laid the foundation for the entire study of black body radiation, making it one of the starting points of modern theoretical physics.

Kirchhoff's law establishes a fundamental relation between the emissive and absorptive properties of a body in thermal equilibrium.

11.3.1 Statement of Kirchhoff's Law

At a given temperature and wavelength, the ratio of the spectral emissive power to the spectral absorptivity is the same for all bodies and is equal to that of a perfect black body. Mathematically,

$$\frac{E_\lambda}{a_\lambda} = E_\lambda^{(b)} \tag{11.15}$$

where E_λ is the spectral emissive power of the body, a_λ is its spectral absorptivity, and $E_\lambda^{(b)}$ is the spectral emissive power of a black body at the same temperature and wavelength.

11.3.2 Implication of the Law

From the above relation, we can write:

$$E_\lambda = a_\lambda \, E_\lambda^{(b)} \tag{11.16}$$

This shows that a good absorber is also a good emitter. In particular, for a perfect black body,

$$a_\lambda = 1 \quad \Rightarrow \quad E_\lambda = E_\lambda^{(b)} \tag{11.17}$$

Thus, a black body is the most efficient emitter of radiation.

11.3.3 Proof

Consider two bodies placed inside an enclosure maintained at a constant temperature T, such that thermal equilibrium is established. Let radiation of wavelength λ be incident on both bodies.

Let:

- a_λ be the spectral absorptivity,
- E_λ be the spectral emissive power.

In thermal equilibrium, the energy absorbed must be equal to the energy emitted; otherwise, the temperature would change.

If the incident radiation has spectral energy density u_λ, then the energy absorbed per unit time is proportional to $a_\lambda u_\lambda$, and the energy emitted per unit time is proportional to E_λ.

Thus, for equilibrium:

$$E_\lambda \propto a_\lambda u_\lambda \tag{11.18}$$

Since u_λ is the same for all bodies in the enclosure at equilibrium, we obtain:

$$\frac{E_\lambda}{a_\lambda} = \text{constant} \tag{11.19}$$

This constant depends only on temperature and wavelength, and is identified as the spectral emissive power of a black body:

$$\frac{E_\lambda}{a_\lambda} = E_\lambda^{(b)} \tag{11.20}$$

11.3.4 Black Body

- Kirchhoff's law shows that emission and absorption are not independent properties.
- It provides a method to define black body radiation independent of the nature of materials.
- It implies that no body can emit more radiation than a black body at the same temperature.
- It forms the basis for the development of Planck's radiation law.

All bodies at a temperature above absolute zero emit electromagnetic radiation due to the thermal motion of their charged particles. This radiation, known as **thermal radiation**, does not require a material medium for its propagation and can occur even in vacuum.

The nature and intensity of radiation emitted by a body depend on its temperature and surface properties. The idealized concept used to study thermal radiation is that of a **blackbody**.

A **blackbody** is an ideal body that absorbs all the radiation incident upon it, regardless of wavelength and direction. Since it absorbs perfectly, it is also the most efficient emitter of radiation.

In practice, a blackbody can be approximated by a cavity with a small hole. Radiation entering the hole undergoes multiple reflections inside the cavity and is almost completely absorbed.

The radiation emitted by a blackbody in thermal equilibrium is called **blackbody radiation**. The distribution of energy with wavelength depends only on the temperature of the blackbody and not on its material composition.

11.4 Wien's Displacement Law

Wilhelm Wien's Legacy

Wilhelm Wien was a German physicist known for his pioneering work on thermal radiation. He formulated Wien's displacement law, which describes how the peak wavelength of black body radiation shifts with temperature.
Wien was awarded the Nobel Prize in Physics in 1911 for his discoveries regarding the laws governing the radiation of heat.

Wien's displacement law describes how the wavelength corresponding to the maximum emission of a black body radiation spectrum depends on temperature.

The wavelength λ_{max} at which the spectral emissive power of a black body is maximum is inversely proportional to its absolute temperature T:

$$\lambda_{\text{max}} T = b \tag{11.21}$$

where b is Wien's displacement constant, given by

$$b \approx 2.898 \times 10^{-3}\,\text{m K} \tag{11.22}$$

As the temperature of a black body increases, the peak of its radiation curve shifts toward shorter wavelengths. This implies:

- At low temperatures, radiation is mainly in the infrared region.
- At higher temperatures, radiation shifts toward the visible region.
- At very high temperatures, radiation extends into the ultraviolet region.

Thus, hotter bodies appear to emit radiation of shorter wavelengths.

- It explains the change in color of hot objects (e.g., red $\rightarrow$ yellow $\rightarrow$ white).
- It is used to estimate the surface temperature of stars.
- It confirms that black body radiation depends only on temperature.

11.5 Stefan–Boltzmann Law

J. Stefan and His student

Josef Stefan, an Austrian physicist, experimentally discovered that the total energy radiated by a black body is proportional to the fourth power of its temperature.
His student, Ludwig Boltzmann, later provided the theoretical derivation of this law using thermodynamics and statistical mechanics.
Boltzmann's work was crucial in establishing the connection between microscopic atomic behavior and macroscopic physical laws, making him one of the founders of statistical mechanics.

The Stefan–Boltzmann law gives the total energy emitted per unit area of a black body across all wavelengths as a function of its temperature.

The total emissive power of a perfect black body is directly proportional to the fourth power of its absolute temperature:

$$E = \sigma T^4 \tag{11.23}$$

where σ is the Stefan–Boltzmann constant, given by

$$\sigma = 5.67 \times 10^{-8}\,\mathrm{W\ m^{-2}K^{-4}} \tag{11.24}$$

For a Real Body

For a real surface, the emissive power is less than that of a black body and is given by:

$$E = e\,\sigma T^4 \tag{11.25}$$

where e is the emissivity of the surface.

Net Radiative Heat Loss

If a body at temperature T is surrounded by an enclosure at temperature T_0, the net rate of heat loss per unit area is:

$$E_{\text{net}} = \sigma\left(T^4 - T_0^4\right) \tag{11.26}$$

For a real body:

$$E_{\text{net}} = e\,\sigma\left(T^4 - T_0^4\right) \tag{11.27}$$

Applications

- Estimation of energy output of stars.
- Calculation of radiative heat loss in engineering systems.
- Determination of surface temperatures in thermal physics.

11.6 Rayleigh–Jeans Law

Lord Rayleigh and Sir James Jeans

Lord Rayleigh (John William Strutt) and Sir James Jeans were British physicists who developed the Rayleigh–Jeans law based on classical physics.
Rayleigh was awarded the Nobel Prize in Physics in 1904 for his discovery of argon.
Jeans made significant contributions to astrophysics and the theory of radiation.

11.6.1 Derivation

The Rayleigh–Jeans law is a classical result that describes the spectral distribution of black body radiation at long wavelengths.

The derivation is based on the following assumptions:

- Radiation inside a cavity is treated as standing electromagnetic waves.

- The number of modes in the frequency range ν to $\nu + d\nu$ is:

$$g(\nu)\, d\nu = \frac{8\pi\nu^2}{c^3}\, d\nu \tag{11.28}$$

- By the equipartition theorem, each mode has an average energy ($\frac{1}{2}k_BT$ per degree of freedom, and Electromagnetic wave has degree of freedom = 2):

$$\langle E \rangle = kT \tag{11.29}$$

Thus, the spectral energy density becomes:

$$\boxed{u_\nu d\nu = g(\nu)\, \langle E \rangle d\nu = \frac{8\pi\nu^2 kT}{c^3} d\nu} \tag{11.30}$$

Limitations

- The law agrees well with experimental results at long wavelengths (low frequencies).
- However, it fails completely at short wavelengths (high frequencies).
- As $\lambda \to 0$, the energy density $u_\lambda \to \infty$, which is physically unacceptable.

11.6.2 Ultraviolet Catastrophe

The failure of the Rayleigh–Jeans law at short wavelengths is known as the **ultraviolet catastrophe**.

According to classical theory:

$$u_\lambda \propto \frac{1}{\lambda^4} \tag{11.31}$$

Thus, as $\lambda \to 0$ (ultraviolet region), the energy density tends to infinity:

$$u_\lambda \to \infty \tag{11.32}$$

This implies that a black body would emit infinite energy in the ultraviolet region, which contradicts experimental observations.

- The ultraviolet catastrophe revealed the failure of classical physics in explaining black body radiation.
- It indicated the need for a new theoretical framework.
- This problem was resolved by Planck, who introduced the concept of energy quantization.

11.7 Planck's Radiation Law

Planck's Legacy

Max Planck was a German physicist and the founder of quantum theory. In 1900, he proposed that energy is not emitted continuously but in discrete packets called quanta.
Using this revolutionary idea, Planck successfully derived the correct formula for black body radiation, resolving the ultraviolet catastrophe.
He was awarded the Nobel Prize in Physics in 1918 for his discovery of energy quanta. Planck's work marked the beginning of modern physics and fundamentally changed our understanding of nature.

11.7.1 Planck's Quantum Hypothesis

Planck made the following fundamental assumptions:

- Energy is not emitted or absorbed continuously, but in discrete packets called **quanta**.
- The energy of each quantum is proportional to the frequency of radiation:

$$E = h\nu \tag{11.33}$$

  where h is Planck's constant.
- The oscillators in the walls of a black body can have only discrete energy values:

$$E_n = nh\nu \quad (n = 0, 1, 2, 3, \dots) \tag{11.34}$$

11.7.2 Average Energy of an Oscillator

According to statistical mechanics, the probability of an oscillator having energy E_n is given by the Boltzmann distribution:

$$P_n \propto e^{-E_n/kT} \tag{11.35}$$

Thus, the average energy is:

$$\langle E \rangle = \frac{\sum_{n=0}^{\infty} E_n e^{-E_n/kT}}{\sum_{n=0}^{\infty} e^{-E_n/kT}} \tag{11.36}$$

Substituting $E_n = nh\nu$:

$$\langle E \rangle = \frac{\sum_{n=0}^{\infty} nh\nu\, e^{-nh\nu/kT}}{\sum_{n=0}^{\infty} e^{-nh\nu/kT}} \tag{11.37}$$

Let

$$x = e^{-h\nu/kT} \tag{11.38}$$

Then,

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}, \quad \sum_{n=0}^{\infty} nx^n = \frac{x}{(1-x)^2} \tag{11.39}$$

Using these results:

$$\langle E \rangle = h\nu \frac{x}{1-x} \tag{11.40}$$

Substituting back $x = e^{-h\nu/kT}$:

$$\langle E \rangle = \frac{h\nu}{e^{h\nu/kT} - 1} \tag{11.41}$$

11.7.3 Planck's Radiation Law

The number of electromagnetic modes per unit volume in the frequency range ν to $\nu + d\nu$ is:

$$g(\nu)\, d\nu = \frac{8\pi\nu^2}{c^3}\, d\nu \tag{11.42}$$

Thus, the spectral energy density is:

$$u_\nu d\nu = g(\nu)\, \langle E \rangle d\nu \tag{11.43}$$

Substituting:

$$u_\nu d\nu = \frac{8\pi\nu^2}{c^3} \cdot \frac{h\nu}{e^{h\nu/kT} - 1} d\nu \tag{11.44}$$

Hence,

$$\boxed{u_\nu d\nu = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/kT} - 1} d\nu} \tag{11.45}$$

This is Planck's radiation law in frequency form.
Using the relation $\nu = \frac{c}{\lambda}$, we obtain:

$$\boxed{u_\lambda d\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1} d\lambda} \tag{11.46}$$

- It successfully explains the entire black body spectrum.
- It resolves the ultraviolet catastrophe.
- It marks the origin of quantum theory.
- It shows that energy exchange is quantized.

11.7.4 Derivation of Classical Laws from Planck's Radiation Law

Planck's radiation law successfully explains black body radiation. It also reduces to previously known laws under appropriate limits. In this section, we derive Wien's law, Rayleigh–Jeans law, and Stefan–Boltzmann law from Planck's formula.

Rayleigh–Jeans Law (Low Frequency Limit)

For low frequencies ($h\nu \ll kT$), we use the approximation:

$$e^{h\nu/kT} \approx 1 + \frac{h\nu}{kT} \tag{11.47}$$

Thus,

$$e^{h\nu/kT} - 1 \approx \frac{h\nu}{kT} \tag{11.48}$$

Substituting into Planck's law:

$$u_\nu d\nu = \frac{8\pi h\nu^3}{c^3} \cdot \frac{kT}{h\nu} d\nu \tag{11.49}$$

Simplifying:

$$u_\nu d\nu = \frac{8\pi\nu^2 kT}{c^3} d\nu \tag{11.50}$$

This is the Rayleigh–Jeans law.

Wien's Law (High Frequency Limit)

For high frequencies ($h\nu \gg kT$), we use:

$$e^{h\nu/kT} \gg 1 \quad \Rightarrow \quad e^{h\nu/kT} - 1 \approx e^{h\nu/kT} \tag{11.51}$$

Substituting into Planck's law:

$$u_\nu d\nu = \frac{8\pi h\nu^3}{c^3} e^{-h\nu/kT} d\nu \tag{11.52}$$

This is Wien's radiation law:

$$u_\nu d\nu \propto \nu^3 e^{-h\nu/kT} d\nu \tag{11.53}$$

Stefan–Boltzmann Law

To derive Stefan's law, we integrate Planck's distribution over all frequencies:

$$u = \int_0^\infty u_\nu \, d\nu \tag{11.54}$$

Substituting Planck's law:

$$u = \int_0^\infty \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/kT} - 1} \, d\nu \tag{11.55}$$

Let

$$x = \frac{h\nu}{kT} \quad \Rightarrow \quad d\nu = \frac{kT}{h} dx \tag{11.56}$$

Then,

$$u = \frac{8\pi (kT)^4}{h^3 c^3} \int_0^\infty \frac{x^3}{e^x - 1} \, dx \tag{11.57}$$

Using the standard result:

$$\int_0^\infty \frac{x^3}{e^x - 1} dx = \frac{\pi^4}{15} \tag{11.58}$$

Thus,

$$u = \frac{8\pi^5 k^4}{15h^3c^3} T^4 \tag{11.59}$$

Since emissive power $E = \frac{c}{4}u$, we get:

$$E = \sigma T^4 \tag{11.60}$$

where

$$\sigma = \frac{2\pi^5 k^4}{15h^3c^2} \tag{11.61}$$

This is the Stefan–Boltzmann law.

Appendix A

The Polytropic Process

A polytropic process is a general thermodynamic path followed by an ideal gas that satisfies the mathematical relationship:

$$PV^x = \text{constant} \tag{A.1}$$

where x is the polytropic index.

By varying the value of x, we recover the four fundamental processes:

Index (x)	**Process Type**	**Mathematical Result**	**Physical Meaning**
$x = 0$	Isobaric	$P = \text{const}$	Constant Pressure
$x = 1$	Isothermal	$PV = \text{const}$	Constant Temperature
$x = \gamma$	Adiabatic	$PV^\gamma = \text{const}$	No heat exchange ($Q = 0$)
$x = \infty$	Isochoric	$V = \text{const}$	Constant Volume

Table A.1: Polytropic process

Appendix B

Equation of States

An equation of state (EOS) relates thermodynamic variables such as pressure, volume, and temperature, and provides insight into the physical behavior of different systems. The following equations are presented in a progression from simple empirical relations to more advanced and specialized models.

Virial Equation of State

The virial equation expresses deviations from ideal gas behavior as a power series expansion.

$$\frac{PV_m}{RT} = 1 + \frac{B(T)}{V_m} + \frac{C(T)}{V_m^2} + \cdots \tag{B.1}$$

Parameters: $B(T)$, $C(T)$: temperature-dependent interaction coefficients.

Berthelot Equation of State

Introduces temperature dependence in the attractive term.

$$P = \frac{RT}{V_m - b} - \frac{a}{TV_m^2} \tag{B.2}$$

Parameters: a: attraction parameter, b: co-volume.

Clausius Equation of State

A refinement including an additional correction parameter.

$$P = \frac{RT}{V_m - b} - \frac{a}{T(V_m + c)^2} \tag{B.3}$$

Parameters: a, b, c: interaction constants.

Redlich–Kwong Equation of State

Improves real gas behavior with temperature-dependent attraction.

$$P = \frac{RT}{V_m - b} - \frac{a}{\sqrt{T}V_m(V_m + b)} \tag{B.4}$$

Parameters: a, b.

Soave–Redlich–Kwong (SRK) Equation

Further refinement using a temperature correction function.

$$P = \frac{RT}{V_m - b} - \frac{a\alpha(T)}{V_m(V_m + b)} \tag{B.5}$$

Parameters: a, b, $\alpha(T)$.

Peng–Robinson Equation of State

Widely used due to improved accuracy near critical conditions.

$$P = \frac{RT}{V_m - b} - \frac{a\alpha(T)}{V_m(V_m + b) + b(V_m - b)} \tag{B.6}$$

Parameters: a, b, $\alpha(T)$.

Hard-Sphere Equation of State

Describes particles with finite size and no attractive interaction.

$$Z = \frac{1 + \eta + \eta^2 - \eta^3}{(1 - \eta)^3} \tag{B.7}$$

Parameters: Z: compressibility factor, η: packing fraction.

Benedict–Webb–Rubin (BWR) Equation

An empirical equation providing high accuracy over a wide range.

$$P = RT\rho + (B_0RT - A_0)\rho^2 + (bRT - a)\rho^3 + a\alpha\rho^6 \tag{B.8}$$

Parameters: A_0, B_0, a, b, α.

Radiation Equation of State

Describes thermodynamic properties of blackbody radiation.

$$P = \frac{1}{3}aT^4 \tag{B.9}$$

Parameter: a: radiation constant.

Degenerate Fermi Gas

Applicable to systems of fermions at low temperatures.

$$P = K\rho^{5/3} \tag{B.10}$$

Parameter: K: constant depending on system properties.

Bose Gas Equation of State

Describes bosonic systems with quantum statistical behavior.

$$P = \frac{kT}{\lambda^3} g_{5/2}(z) \tag{B.11}$$

Parameters: k, λ, $g_{5/2}(z)$, z.

Appendix C

Spherical Coordinate System

Definition of Spherical Coordinates

In the spherical coordinate system, the position of a point in space is specified by the coordinates (r, θ, ϕ), where:

- r is the radial distance from the origin,
- θ is the polar angle measured from the positive z-axis ($0 \leq \theta \leq \pi$),
- ϕ is the azimuthal angle measured in the xy-plane from the positive x-axis ($0 \leq \phi < 2\pi$).

The transformation between Cartesian and spherical coordinates is given by

$$x = r \sin\theta \cos\phi, \tag{C.1}$$
$$y = r \sin\theta \sin\phi, \tag{C.2}$$
$$z = r \cos\theta. \tag{C.3}$$

Conversely,

$$r = \sqrt{x^2 + y^2 + z^2}, \tag{C.4}$$
$$\theta = \cos^{-1}\left(\frac{z}{r}\right), \tag{C.5}$$
$$\phi = \tan^{-1}\left(\frac{y}{x}\right). \tag{C.6}$$

Unit Vectors in Spherical Coordinates

The spherical coordinate system uses three mutually orthogonal unit vectors:

- $\hat{\mathbf{r}}$ in the radial direction,
- $\hat{\boldsymbol{\theta}}$ in the direction of increasing θ,
- $\hat{\boldsymbol{\phi}}$ in the direction of increasing ϕ.

These unit vectors vary with position and are related to the Cartesian unit vectors by

$$\hat{\mathbf{r}} = \sin\theta \cos\phi\, \hat{\mathbf{x}} + \sin\theta \sin\phi\, \hat{\mathbf{y}} + \cos\theta\, \hat{\mathbf{z}}, \tag{C.7}$$
$$\hat{\boldsymbol{\theta}} = \cos\theta \cos\phi\, \hat{\mathbf{x}} + \cos\theta \sin\phi\, \hat{\mathbf{y}} - \sin\theta\, \hat{\mathbf{z}}, \tag{C.8}$$
$$\hat{\boldsymbol{\phi}} = -\sin\phi\, \hat{\mathbf{x}} + \cos\phi\, \hat{\mathbf{y}}. \tag{C.9}$$

Differential Length, Area, and Volume Elements

In spherical coordinates, the differential displacement vector is

$$d\mathbf{l} = dr\,\hat{\mathbf{r}} + r\,d\theta\,\hat{\boldsymbol{\theta}} + r\sin\theta\,d\phi\,\hat{\boldsymbol{\phi}}. \tag{C.10}$$

The corresponding differential area elements are

$$d\mathbf{a}_r = r^2\sin\theta\,d\theta\,d\phi\,\hat{\mathbf{r}}, \tag{C.11}$$

$$d\mathbf{a}_\theta = r\sin\theta\,dr\,d\phi\,\hat{\boldsymbol{\theta}}, \tag{C.12}$$

$$d\mathbf{a}_\phi = r\,dr\,d\theta\,\hat{\boldsymbol{\phi}}. \tag{C.13}$$

The differential volume element is

$$d\tau = r^2\sin\theta\,dr\,d\theta\,d\phi. \tag{C.14}$$

Gradient, Divergence, and Curl in Spherical Coordinates

For a scalar field $\phi(r,\theta,\phi)$, the gradient is

$$\boldsymbol{\nabla}\phi = \hat{\mathbf{r}}\frac{\partial\phi}{\partial r} + \hat{\boldsymbol{\theta}}\frac{1}{r}\frac{\partial\phi}{\partial\theta} + \hat{\boldsymbol{\phi}}\frac{1}{r\sin\theta}\frac{\partial\phi}{\partial\phi}. \tag{C.15}$$

For a vector field $\mathbf{A} = A_r\hat{\mathbf{r}} + A_\theta\hat{\boldsymbol{\theta}} + A_\phi\hat{\boldsymbol{\phi}}$, the divergence is

$$\boldsymbol{\nabla}\cdot\mathbf{A} = \frac{1}{r^2}\frac{\partial}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta} + \frac{1}{r\sin\theta}\frac{\partial A_\phi}{\partial\phi}. \tag{C.16}$$

The curl of $\mathbf{A}$ is given by

$$\boldsymbol{\nabla}\times\mathbf{A} = \frac{1}{r\sin\theta}\begin{vmatrix} \hat{\mathbf{r}} & r\hat{\boldsymbol{\theta}} & r\sin\theta\,\hat{\boldsymbol{\phi}} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial\theta} & \frac{\partial}{\partial\phi} \\ A_r & rA_\theta & r\sin\theta A_\phi \end{vmatrix}. \tag{C.17}$$

Laplacian in Spherical Coordinates

The Laplacian of a scalar field $\phi(r,\theta,\phi)$ in spherical coordinates is

$$\nabla^2\phi = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\phi}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\phi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\phi}{\partial\phi^2}. \tag{C.18}$$

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